

Astronomy Question Bank

These questions were carefully written or selected to prepare students for the astronomy team selection process in Brazil. Some questions are not original; those are marked with () before the statement. The question-bank template is the same one used by Professor Kevin Zhou. His work is invaluable, and many ideas in this set can be found in his handouts.

1 Celestial Mechanics

[5] **Problem 1.** Iaum and Sevla, two renowned physicists, intend to launch a space probe to explore the limits of Newtonian physics and the relativistic corrections required near a black hole. They need your help to study the *effective potential* of such a body.

a) The energy of a body of mass m orbiting a massive body of mass M can be written as:

$$\frac{m\dot{r}^2}{2} + V_{eff}(r)$$

where $\dot{r}$ is the radial velocity of the body. Find $V_{eff}(r)$ in terms of G , M , L , m , and r .

b) According to Newtonian physics, what is the smallest radius at which a body can have a circular orbit around a black hole?

c) What is the frequency of small radial oscillations, ω_r , about this radius?

The gravitational potential near these bodies must include a relativistic correction and has the form:

$$V(r) = -\frac{GMm}{r} - \frac{GML^2}{mc^2r^3}$$

d) Explain why this correction allows a particle to fall into the center of a black hole, $r = 0$, and why this is impossible in Newtonian physics.

e) What is the smallest value of L for which a particle can orbit the black hole on a circular orbit? What is the value of the radius in this case?

f) Assuming $\lim_{L \rightarrow \infty}$, find the smallest distance to which a particle on an open orbit can approach a black hole.

Solution

a) In the classical form, we have:

$$E = \frac{mv^2}{2} - \frac{GMm}{r}$$

Note that we can write $v^2 = v_\theta^2 + v_r^2$, where v_θ and v_r are, respectively, the tangential and radial velocities of the body. Since $v_r = \dot{r}$, we need an expression that relates v_θ to another quantity. The best way to make this relation is by using the angular momentum, since:

$$L = mrv_\theta \rightarrow v_\theta = \frac{L}{mr}$$

Substituting into the energy expression, we have:

$$E = \frac{m\dot{r}^2}{2} + \frac{L^2}{2mr^2} - \frac{GMm}{r}$$

Since we want an answer of the form $E = m\dot{r}^2/2 + V_{eff}(r)$, by analogy we have:

$$V_{eff}(r) = -\frac{GMm}{r} + \frac{L^2}{2mr^2}$$

b) The greatest limiting speed of a body is c , disregarding relativistic effects:

$$\frac{mc^2}{2} - \frac{GMm}{R} = -\frac{GMm}{2R}$$

Isolating R :

$$R = \frac{GM}{c^2}$$

c) For small oscillations, we have the approximation:

$$\omega_r \approx \sqrt{\frac{V''_{eff}(R)}{m}}$$

Calculating:

$$V'_{eff}(R) = \frac{GMm}{r^2} - \frac{L^2}{mr^3}$$

$$V''_{eff}(R) = -\frac{2GMm}{r^3} + \frac{3L^2}{mr^4}$$

Using $L = mvr$ (circular orbits) and $v \equiv c$, we arrive at:

$$V''_{eff}(R) = \frac{mc^6}{G^2M^2}$$

With this, we can conclude that:

$$\omega_r = \frac{c^3}{GM}$$

d) When we take the limit $\lim_{r \rightarrow 0} V_{eff}(r)$, in Newtonian physics the dominant term is $L^2/2mr^2$, that is, an energy of $+\infty$ would be required to reach the center of the black hole. However, after the relativistic correction, the dominant term is $-GML^2/mc^2r^3$, that is, an energy of $-\infty$ would be required to reach the center of the black hole. That is why it is not possible to reach the center of a black hole in Newtonian physics.

e) On a circular orbit, r is constant, which implies that the force in the radial direction vanishes, that is:

$$V'_{eff}(r) = 0$$

Calculating:

$$V'_{eff}(r) = \frac{GMm}{r^2} - \frac{L^2}{mr^3} + \frac{3GML^2}{mc^2r^4} = 0$$

Simplifying, we have:

$$r^2 - \frac{L^2}{GMm^2}r + \frac{3L^2}{m^2c^2} = 0$$

Note that the solution of this equation must be real, since r cannot be imaginary, which imposes the condition:

$$\frac{L^4}{G^2M^2m^4} - \frac{12L^2}{m^2c^2} \geq 0$$

In the equality case, the value of L that satisfies this equation is:

$$L = \frac{\sqrt{12GMm}}{c}$$

The value of r in this case is:

$$r = \frac{L^2}{2GMm^2} = r_{min} = \frac{6GM}{c^2}$$

f) The complete expression for r is:

$$r = \frac{\frac{L^2}{GMm^2} \pm \sqrt{\left(\frac{L^2}{GMm^2}\right)^2 - \frac{12L^2}{m^2c^2}}}{2} = \frac{L^2 - \sqrt{L^4 - 12(GMmL/c)^2}}{2GMm^2}$$

In the limit $L \rightarrow \infty$, we have:

$$r_{min} = \lim_{L \rightarrow \infty} \frac{L^2 - \sqrt{L^4 - 12(GMmL/c)^2}}{2GMm^2} = \frac{6(GMm/c)^2}{2(GMm^2)} = \frac{3GM}{c^2}$$

[3] **Problem 2.** An important point in the study of orbits is what happens when the potential follows something unusual. The goal of this question is to study that phenomenon. Consider the angular momentum L and the mass of the body m .

- For a general orbit that has a potential $V(r)$ of the form ψr^k , find the value r_0 of a circular orbit for this potential.
- Find the frequency of small oscillations, ω_r , about this radius.
- Let ω_θ be the angular frequency of the orbit. Find the value of the ratio $\frac{\omega_r}{\omega_\theta}$. An orbit can be closed only if this ratio is an integer. Why? Find values of k for which the orbit is closed.

Solution

a) Since on a circular orbit $V'_{eff}(r) = 0$, we have:

$$V'_{eff}(r) = -\frac{L^2}{mr^3} + k\psi r^{k-1} = 0$$

$$k\psi r^{k-1} = \frac{L^2}{mr^3} \rightarrow r_0 = \left(\frac{L^2}{k\psi m} \right)^{\frac{1}{k+2}}$$

b) Using $\omega_r^2 = V''_{eff}(r_0)/m$:

$$V''_{eff}(r) = \frac{3L^2}{mr^4} + k(k-1)\psi r^{k-2}$$

$$V''_{eff}(r) = r^{-4} \left(\frac{3L^2}{m} + k(k-1)\psi r^{k+2} \right)$$

$$V''_{eff}(r) = r^{-4} \left(\frac{3L^2}{m} + k(k-1)\psi \frac{L^2}{k\psi m} \right)$$

$$V''_{eff}(r) = \frac{L^2}{r_0^4 m} (k+2)$$

Thus, we have:

$$\omega_r = \frac{L}{r_0^2 m} \sqrt{k+2}$$

c) By the definition of L , we have:

$$L = mr_0^2 \omega_\theta \rightarrow \omega_\theta = \frac{L}{mr_0^2}$$

Therefore:

$$\frac{\omega_r}{\omega_\theta} = \sqrt{k+2}$$

Note that, for the orbit to be a closed figure, this ratio must be an integer, so $k+2$ must be a perfect square. Any value $k = n^2 - 2 \forall n \in \mathbb{N}$ satisfies this equation.

[4] **Problem 3.** The hodograph is one of the lesser-known ways of solving celestial-mechanics problems. Using this device, one can solve complex problems (involving, for example, finding orbital parameters, velocity parameters, and how to minimize the eccentricity of an orbit). Although the hodograph has many uses, the goal of this exercise is to prove the Hodograph Theorem, which states that, on closed orbits, the velocity forms a circle in vector space. We will prove this theorem in two ways.

Method 1)

- I. Starting from Newton's second law and the conservation of angular momentum, find an expression for $d\mathbf{v}/d\theta$. Leave your answer in terms of the angular momentum per unit mass, h , and $\mu = GM$.
- II. Where is the center of this circle located? Taking the central body at the origin of the system, determine an expression for the distance between the central body and the center of the hodograph.

Method 2)

Method 2 uses more advanced notions of calculus and starts from the conservation of the *Laplace-Runge-Lenz vector*, $\mathbf{A}$.

I. The vector $\mathbf{A}$ is given by the expression:

$$\mathbf{A} = \mathbf{p} \times \mathbf{L} - GMm^2 \hat{\mathbf{r}}$$

The first step of the proof is to show that $\mathbf{A}$ is constant. To do so, differentiate it with respect to time and draw your own conclusions.

II. Since the vector $\mathbf{L}$ is constant, $\mathbf{A} \times \mathbf{L}$ is also a constant. Using the vector identity $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$, prove that $\mathbf{v}$ forms a circle in vector space.

Solution**Method 1)**

I. Newton's second law tells us that:

$$-\frac{GMm}{r^2} \hat{\mathbf{r}} = m \frac{d\mathbf{v}}{dt}$$

By the definition of angular momentum:

$$L = mr^2 \frac{d\theta}{dt} \rightarrow \therefore dt = \frac{mr^2}{L} d\theta$$

Substituting dt into Newton's second law:

$$-\frac{GMm}{r^2} \hat{\mathbf{r}} = m \left(\frac{L}{mr^2} \right) \frac{d\mathbf{v}}{d\theta}$$

Isolating $d\mathbf{v}/d\theta$, we have:

$$\boxed{\frac{d\mathbf{v}}{d\theta} = -\frac{GMm}{L} \hat{\mathbf{r}} \equiv -\frac{\mu}{h} \hat{\mathbf{r}}}$$

Note that this expression is a circle in vector space and has radius $R = \mu/h$, since $d\mathbf{v}/d\theta$ is constant and points in the radial direction.

II. As we saw in the previous item, the hodograph forms a circle of radius $R = \mu/h$. Use the following figure as support:

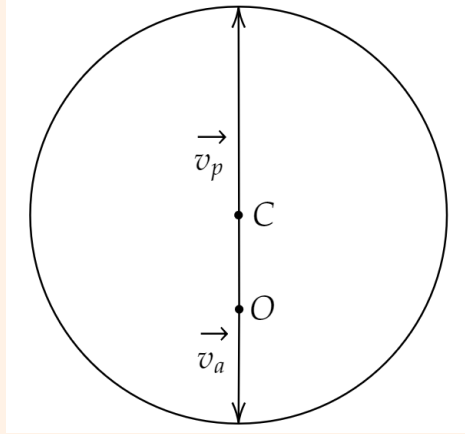


Figure 1: Diagram of the hodograph

The point O represents the origin of the system (location of the central body). Note that the smallest distance between O and the hodograph must correspond to the smallest speed (apastron) and the largest distance must correspond to the largest speed (periastron). Also note that $v_a + \overline{OC} = v_p - \overline{OC}$. Setting up the equations, we have:

$$v_a + v_p = 2R \equiv \frac{2\mu}{h}$$

$$v_p - v_a = 2\overline{OC}$$

We can write v_p and v_a in terms of the angular momentum:

$$L = ma(1 - e)v_p \rightarrow v_p = \frac{h}{a(1 - e)}$$

$$L = ma(1 + e)v_a \rightarrow v_a = \frac{h}{a(1 + e)}$$

Substituting into the first equation:

$$\frac{h}{a} \left(\frac{1}{1 + e} + \frac{1}{1 - e} \right) = \frac{2\mu}{h}$$

Here, we can isolate a :

$$\frac{h}{a} \left(\frac{2}{(1 - e)^2} \right) = \frac{2\mu}{h}$$

$$a = \frac{h^2}{\mu} \frac{1}{(1 - e)^2}$$

With this in mind, we can move on to the second expression:

$$\frac{h}{a} \left(\frac{1}{1 - e} - \frac{1}{1 + e} \right) = 2\overline{OC}$$

$$h \left(\frac{\mu(1-e)^2}{h^2} \right) \left(\frac{2e}{(1-e)^2} \right) = 2\overline{OC}$$

Finally, solving for $\overline{OC}$, we have:

$$\boxed{\overline{OC} = \frac{\mu}{h}e}$$

For an ellipse ($e < 1$), the origin lies inside the circle; for a parabola ($e = 1$), the origin lies on the boundary of the circle; and for a hyperbola ($e > 1$), the origin lies outside the circle.

Method 2

I. Differentiating $\mathbf{A}$ with respect to time, we have:

$$\frac{d\mathbf{A}}{dt} = \frac{d}{dt}(\mathbf{p} \times \mathbf{L}) - GMm^2 \frac{d}{dt} \hat{\mathbf{r}}$$

When we differentiate a *unit vector* with respect to time, we have:

$$\frac{d\hat{\mathbf{r}}}{dt} = \boldsymbol{\omega} \times \hat{\mathbf{r}}$$

where $\boldsymbol{\omega}$ is the velocity with which the unit vector translates. Thus:

$$\frac{d\mathbf{A}}{dt} = \left(\frac{d\mathbf{p}}{dt} \right) \times \mathbf{L} + \mathbf{p} \times \left(\frac{d\mathbf{L}}{dt} \right) - GMm^2(\boldsymbol{\omega} \times \hat{\mathbf{r}})$$

Now we have a few considerations. First, note that $d\mathbf{L}/dt = 0$, since there are only central forces on the orbit. Second, $d\mathbf{p}/dt = \mathbf{F}$. Furthermore, we have $\mathbf{L} = mr^2\boldsymbol{\omega}$. Returning to the equation:

$$\begin{aligned} \frac{d\mathbf{A}}{dt} &= -\frac{GMm}{r^2} \hat{\mathbf{r}} \times (mr^2\boldsymbol{\omega}) - GMm^2(\boldsymbol{\omega} \times \hat{\mathbf{r}}) \\ \frac{d\mathbf{A}}{dt} &= -GMm^2(\hat{\mathbf{r}} \times \boldsymbol{\omega}) - GMm^2(\boldsymbol{\omega} \times \hat{\mathbf{r}}) \end{aligned}$$

For any two vectors, $\mathbf{i}$ and $\mathbf{j}$, it holds that:

$$\mathbf{i} \times \mathbf{j} = -\mathbf{j} \times \mathbf{i}$$

Using this property, we can conclude that:

$$\boxed{\frac{d\mathbf{A}}{dt} = 0}$$

Therefore, $\mathbf{A}$ is constant!

II. Writing the expression and using *BAC-CAB*, we have:

$$\mathbf{A} \times \mathbf{L} = (\mathbf{p} \times \mathbf{L}) \times \mathbf{L} - GMm^2 \hat{\mathbf{r}} \times \mathbf{L}$$

$$(\mathbf{A} + GMm^2 \hat{\mathbf{r}}) \times \mathbf{L} = -\mathbf{L} \times (\mathbf{p} \times \mathbf{L}) = \mathbf{p}(-\mathbf{L} \cdot \mathbf{L}) - \mathbf{L}(-\mathbf{L} \cdot \mathbf{p})$$

Since $\mathbf{L}$ and $\mathbf{p}$ are always perpendicular, $\mathbf{L} \cdot \mathbf{p} = 0$, so:

$$(\mathbf{A} + GMm^2 \hat{\mathbf{r}}) \times \mathbf{L} = -mL^2 \mathbf{v}$$

Since both $\mathbf{A}$ and $\hat{\mathbf{r}}$ are perpendicular to $\mathbf{L}$ and constant, $\mathbf{A} + GMm^2 \hat{\mathbf{r}}$ traces a circle. The cross product with $\mathbf{L}$ merely rotates the circle by 90° and multiplies its radius by $\mathbf{L}$. Therefore, $\mathbf{v}$ traces a circle in vector space.

[2] **Problem 4.** One of the most fascinating phenomena of the Solar System is neutron stars. These are very small, extremely dense bodies that spin very fast. In this question, we will build a theoretical model for these stars.

- a) The Vela Pulsar, a neutron star located in the constellation Vela, has a frequency of 11Hz, that is, it spins about its own axis 11 times per second, with equatorial radius $R_e = 9.6$ km and mass $M = 1.88M_\odot$. Given this, calculate the ratio of its equatorial radius to its polar radius.
- b) What is the smallest rotation period the Vela Pulsar can have so that it does not break apart?
- c) Currently, the rate of change of the period of the Vela Pulsar is $\frac{\Delta P}{\Delta t} = 1.25 \cdot 10^{-13}$ s/s. This causes the pulsar to release an enormous amount of energy. The surface temperature of the pulsar is $T \sim 10^6$ K. Calculate the order of magnitude of the ratio between the power emitted due to the increase of the period and the power emitted by the Stefan-Boltzmann law.

Solution

- a) A star is always in hydrostatic equilibrium, so, writing the balance for a point on the equator and for one of the poles,

$$-\frac{GM}{r_e} - \frac{1}{2}\omega^2 r_e^2 = -\frac{GM}{r_p}$$

Using $\omega = 2\pi f$,

$$\frac{GM}{r_p} = \frac{GM}{r_e} + 2\pi^2 f^2 r_e^2$$

$$\boxed{\frac{r_e}{r_p} = 1 + \frac{2\pi^2 f^2 r_e^3}{GM} \approx 1 + 8.4 \cdot 10^{-6}}$$

That is, a neutron star, even spinning absurdly fast, is still practically a perfect sphere!

b) In the limiting situation in which it remains intact, we have:

$$\frac{GM}{r_e^2} = \omega^2 r_e$$

$$\omega^2 = \frac{GM}{r_e^3} \rightarrow T_{min} = \frac{2\pi r_e^3}{GM} = 2.2 \cdot 10^{-8} \text{ s}$$

c) The energy due to rotation has the form:

$$E = \frac{1}{2} I \omega^2$$

where I is the moment of inertia of the star ($I = \frac{2}{5} M r^2$). Therefore, the dissipated power is:

$$P = \frac{dE}{dt} = I \omega \frac{d\omega}{dt}$$

Since $\omega = 2\pi/T$,

$$\frac{d\omega}{dT} = -\frac{2\pi}{T^2} \rightarrow d\omega = -\frac{2\pi}{T^2} dT$$

The expression for the power as a function of the period becomes:

$$P = -\frac{8\pi^2 M r_e^2}{5T^3} \frac{dT}{dt} = -9.1 \cdot 10^{29} \text{ W}$$

Note that, by conservation of energy, all of this power must be converted into luminosity, so $L_{rot} = -P = 9.1 \cdot 10^{29} \text{ W}$. The power coming from thermal radiation is given by:

$$L_{rad} = 4\pi R^2 \sigma T^4 = 6.56 \cdot 10^{25} \text{ W}$$

With this, we can conclude that 99.99% of its luminosity comes from the rotation.

[3] **Problem 5.** (Adapted from PPP) Geométrio, Paulinho, and Hirata love chocolate. One day, they were exploring a galaxy when they came across a giant ball of chocolate. The three quickly began to eat the ball, first making a straight line from the point P to the center of the ball at O (diagram 1). After that, Hirata falls, colliding at the point O , without braking along the way. Surprisingly, he remains alive and is rescued by Geométrio and Paulinho. After that, still hungry, they continue to eat the giant chocolate sphere and leave a spherical hole of diameter $\overline{PO}$ (diagram 2). Hirata is very clumsy and ends up falling again, starting from the point P and going to O once more.

- a) Find the ratio of Hirata's impact speeds in cases 1 and 2.
- b) Find the ratio of Hirata's fall times in cases 1 and 2.

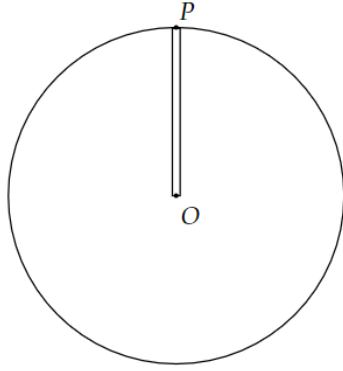


Figure 2: Diagram 1

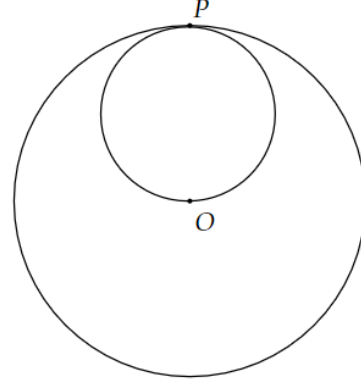


Figure 2: Diagram 2

Solution

- a) The gravity inside a planet is given by:

$$\oint \mathbf{g} \cdot d\mathbf{S} = -4\pi G M_{int}$$

$$4\pi r^2 g = -4\pi G \rho \frac{4\pi}{3} r^3$$

That is, for case 1, $g \propto -\rho r \rightarrow g = -k\rho r$. Using Newton's second law:

$$F = mv \frac{dv}{dr} = -mkr$$

Integrating:

$$\int_0^{v_1} v' dv' = -k\rho \int_0^R r dr$$

$$\frac{v_1^2}{2} = k\rho \frac{R^2}{2} \rightarrow \boxed{v_1 = R\sqrt{k\rho}}$$

Let $\mathbf{j}$ be the vector from the center of the planet to the center of the cavity. Using superposition:

$$\mathbf{a} = -k\rho \mathbf{r} - k(-\rho)(\mathbf{r} - \mathbf{j}) = -k\rho \mathbf{j}$$

That is, in case 2, the acceleration is constant and has magnitude $|a| = \frac{k\rho R}{2}$.

Using Torricelli:

$$v_2^2 = 2aRk\rho R^2 \rightarrow \boxed{v_2 = R\sqrt{k\rho}}$$

Therefore, $v_1/v_2 = 1$.

- b) In the first case, note that we have simple harmonic motion, since the force is proportional to r . The time to reach the bottom equals 1/4 of the period:

$$\ddot{r} = -k\rho r \rightarrow \omega^2 = k\rho \rightarrow T = \frac{2\pi}{\sqrt{k\rho}} \rightarrow t_1 = \frac{T}{4} = \frac{\pi}{2\sqrt{k\rho}}$$

In the second case, we have a constant acceleration, so:

$$t_2 = \frac{v_2}{a_2} = \frac{2R\sqrt{k\rho}}{k\rho R} = \frac{2}{\sqrt{k\rho}}$$

Therefore, the ratio t_1/t_2 is given by:

$$\frac{t_1}{t_2} = \frac{\pi}{4}$$

[5] **Problem 6.** Calculating escape speeds in certain situations can be more complicated than it seems. One example is calculating the launch speed a rocket needs in order to reach a given location.

Hint: The following problems require you to think carefully about the most appropriate reference frame. Do not underestimate the question.

Data: A body can orbit the surface of the Earth with speed $v_0 = 7.9$ km/s, the orbital speed of the Earth around the Sun is $u_0 = 29.7$ km/s, and the radius of the Earth is negligible compared with the Earth-Sun distance. Also assume that, upon leaving the Earth's gravitational field, the distance between the probe and the Sun is the same as the distance between the Earth and the Sun.

- What is the smallest launch speed a rocket needs in order to reach the Sun, assuming that it applies only one impulse? (To check, $v = 31.8$ km/s)
- What is the smallest launch speed a rocket needs in order to escape the Solar System? (To check, $v = 16.7$ km/s)
- How does the answer to item a) change if the rocket can apply a second, very small impulse at some point of its orbit?

Solution

- a) For this to happen, after leaving the Earth, in the Sun's frame the probe must be at rest and, consequently, in the Earth's frame it must have velocity $-u_0$. Using conservation of energy:

$$\begin{aligned} \frac{mv^2}{2} - \frac{GM_{\oplus}m}{R_{\oplus}} &= \frac{mu_0^2}{2} \\ v^2 &= u_0^2 + \frac{2GM_{\oplus}}{R_{\oplus}} \end{aligned}$$

Note that the speed v_0 can also be obtained by equating the centripetal acceleration to the gravitational acceleration:

$$\frac{mv_0^2}{R_{\oplus}} = \frac{GM_{\oplus}m}{R_{\oplus}^2} \rightarrow v_0^2 = \frac{GM_{\oplus}}{R_{\oplus}}$$

Substituting,

$$v^2 = u_0^2 + 2v_0^2 \rightarrow \boxed{v = 31.8 \text{ km/s}}$$

- b) In this case, upon leaving the Earth's gravitational field, in the Sun's frame the body must have speed $\sqrt{2}u_0$, so in the Earth's frame it must have $(\sqrt{2} - 1)u_0$. Again, by conservation of energy:

$$\frac{mv^2}{2} - \frac{GM_{\oplus}m}{R_{\oplus}} = \frac{m(\sqrt{2} - 1)^2 u_0^2}{2}$$

$$v^2 = (\sqrt{2} - 1)^2 u_0^2 + 2v_0^2 \rightarrow \boxed{v = 16.7 \text{ km/s}}$$

- c) Imagine that we send the probe to infinity, that is, we launch it with speed $v = 16.7 \text{ km/s}$, and after that we apply an infinitesimal impulse toward the Sun. This makes the probe travel all the way back and, consequently, reach the Sun.

[3] **Problem 7.** (Kevin Zhou) The rocket equation is given by:

$$v = u \ln \left(\frac{M_0}{M} \right)$$

where v is the speed of the rocket, u the relative speed at which the rocket ejects fuel, M_0 the initial mass of the rocket, and M its current mass.

- Derive the rocket equation starting from conservation of momentum.
- Assuming u is fixed throughout the rocket's trip, what must the value of u be so that the rocket goes from 0 to v using the least possible fuel? (You will need to solve numerically for v/u)
- How must the rocket equation be corrected for a region of space with a constant gravitational field, $\mathbf{g}$, pointing opposite the rocket's motion? Take $\eta = dm/dt = \text{constant}$.

Solution

- a) Consider that the rocket has mass M and velocity v . It ejects fuel at a relative speed u . Let dm be the mass of ejected gas:

$$dp = (v - u)dm$$

But, by definition,

$$dp = vdm + m dv$$

Combining these equations:

$$m dv = -u dm$$

Integrating,

$$\ln m = -\frac{v}{u} + C$$

We know that when $m = M_0$, we have $v = 0$, so $C = \ln M_0$. Thus:

$$v = u \ln \left(\frac{M_0}{M} \right)$$

As expected.

b) The kinetic energy of the gas is given by:

$$K_g = \frac{1}{2} M_g u^2 = \frac{(M_0 - M)u^2}{2}$$

We can find an expression for M using the rocket equation:

$$\frac{M_0}{M} = e^{v/u} \rightarrow M_0 = M e^{v/u}$$

Returning to the previous formula:

$$K_g = \frac{M(e^{v/u} - 1)u^2}{2}$$

Here, M is equivalent to the “structural” mass of the rocket.

To minimize the amount of energy spent on fuel, we have:

$$\frac{dK_g}{du} = 0 \rightarrow -e^{v/u}v + 2u(e^{v/u} - 1) = 0$$

Defining $x = v/u$,

$$e^x x = 2(e^x - 1) \rightarrow x = 2(1 - e^{-x})$$

Solving by iteration, we find:

$$x \approx 1.59 \rightarrow \boxed{u \approx \frac{v}{1.59}}$$

c) In this case, we have:

$$dp = (v - u)dm - mgdt = m dv + v dm$$

$$\frac{dm}{m} = -\frac{dv}{u} - \frac{g}{u} dt$$

Integrating,

$$\ln \left(\frac{M}{M_0} \right) = -\frac{v}{u} - \frac{gt}{u}$$

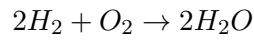
Since η is constant, $M = M_0 - \eta t \rightarrow t = (M_0 - M)/\eta$.

$$\ln \left(\frac{M}{M_0} \right) = -\frac{v}{u} - \frac{g}{u\eta}(M_0 - M)$$

Solving for v ,

$$v = u \ln \left(\frac{M_0}{M} \right) - \frac{g}{\eta}(M_0 - M)$$

- [4] **Problem 8.** One of the most common rocket propellants is a mixture of liquid hydrogen and liquid oxygen. When it begins to burn, the following chemical reaction occurs:



For each mole of hydrogen, this reaction releases 241.8 kJ of energy. Throughout the question, assume that all of this energy is used to move the rocket.

- a) A space expedition is to be carried out in such a way that a Hohmann transfer is required to launch a rocket from Earth to Mars. Calculate the total velocity change Δv needed to perform this maneuver.
- b) From the Δv calculated above, estimate how many tons of propellant must be used to perform this maneuver for a rocket that ejects propellant at speed $u = 3.0$ km/s and has structural mass $M = 140 \cdot 10^3$ kg.

Solution

- a) For the first impulse, we have:

$$v_{0,1}^2 = \frac{GM}{R_T}$$

$$v_{f,1}^2 = GM \left(\frac{2}{R_T} - \frac{1}{a} \right)$$

The transfer orbit has $a = (R_T + R_M)/2$,

$$v_{f,1}^2 = \frac{2GM}{R_T} \left(\frac{R_M}{R_T + R_M} \right)$$

For the second impulse,

$$v_{0,2}^2 = GM \left(\frac{2}{R_M} - \frac{1}{a} \right) = \frac{2GM}{R_M} \left(\frac{R_T}{R_T + R_M} \right)$$

$$v_{f,2}^2 = \frac{GM}{R_M}$$

With $\Delta v = \Delta v_1 + \Delta v_2 = (v_{f,1} - v_{0,1}) + (v_{f,2} - v_{0,1})$, we can write:

$$\Delta v = \sqrt{\frac{2GM}{R_T} \left(\frac{R_M}{R_T + R_M} \right)} - \sqrt{\frac{GM}{R_T}} + \sqrt{\frac{GM}{R_M}} - \sqrt{\frac{2GM}{R_M} \left(\frac{R_T}{R_T + R_M} \right)}$$

Using $M = M_\odot = 2 \cdot 10^{30}$ kg, $R_M = 1.5$ AU, and $R_T = 1$ AU, we find the numerical value $\Delta v \approx 5.42$ km/s.

b) Using the rocket equation, we have:

$$v = u \ln \left(\frac{M_0}{M} \right) \rightarrow \frac{M_0}{M} = 1 + \frac{\mu}{M} = e^{v/u}$$

where μ is the mass of gas.

$$\mu = M(e^{v/u} - 1) \approx 7.12 \cdot 10^5 \text{ kg}$$

This is consistent with reality, since the fuel mass is $\approx 83\%$ of the total mass of the rocket.

- [3] **Problem 9.** Marisso was studying a binary system with inclination i . He found that the largest redshift coming from star 1 was z_1 . Given this, find an expression for the mass of star 2, m_2 . Leave your answer in terms of z_1 , the binary period P , the inclination i , the mass ratio $\lambda = m_2/m_1$, and fundamental constants. Assume circular orbits.

Solution

The speed at which 1 orbits the CM is:

$$v_1 = \frac{2\pi a_1}{P}$$

By geometry, the radial velocity is $v_{r,1} = v_1 \sin i$, so:

$$z_1 \approx \frac{v_{r,1}}{c} = \frac{2\pi a_1 \sin i}{Pc}$$

From the center-of-mass theorem:

$$m_1 a_1 = m_2 a_2 \rightarrow a_2 = \frac{a_1}{\lambda}$$

Using Kepler's third law:

$$\frac{(a_1 + a_2)^3}{P^2} = \frac{G(m_1 + m_2)}{4\pi^2}$$

$$\frac{a_1^3(1 + 1/\lambda)^3}{P^2} = \frac{Gm_2(1 + 1/\lambda)}{4\pi^2}$$

Isolating m_2 :

$$m_2 = \frac{4\pi^2 a_1^3 (1 + 1/\lambda)^2}{GP^2}$$

a_1 can be obtained by evaluating the redshift formula, $a_1 = Pc z_1 / 2\pi \sin i$:

$$m_2 = \frac{4\pi^2}{GP^2} \left(\frac{z_1 Pc}{2\pi \sin i} \right)^3 (1 + 1/\lambda)^2$$

Finally, simplifying:

$$m_2 = \frac{z_1^3 c^3 P}{2\pi G \sin^3 i} (1 + 1/\lambda)^2$$

- [2] **Problem 10.** What is the ratio between a) the gravitational forces caused by the Sun and by the Moon on the surface of the Earth? And b) the tidal forces caused by the same?

Solution

a) The gravitational force has the form:

$$F_G = \frac{GMm}{d^2}$$

Thus,

$$\frac{F_\odot}{F_L} = \frac{M_\odot}{M_L} \left(\frac{d_{T-L}}{d_{T-\odot}} \right)^2 \boxed{\approx 178.33}$$

b) For the tidal forces, we have:

$$F_M \propto \frac{m}{d^3}$$

$$\therefore \frac{F_\odot}{F_L} = \frac{M_\odot}{M_L} \left(\frac{d_{T-L}}{d_{T-\odot}} \right)^3 \boxed{\approx 0.456}$$

- [4] **Problem 11.** (Morin 7.7) A particle of mass m travels on a hyperbolic orbit with a mass M fixed at one of the foci. The speed at infinity is v_0 and the impact parameter is b .

a) Show that the deflection angle of the particle is given by:

$$\phi = \pi - 2 \tan^{-1}(\gamma b)$$

where $\gamma \equiv v_0^2/GM$.

b) Let $d\sigma$ be the cross section of the particle (measured when it is at infinity) that is deflected into a solid angle of size $d\Omega$. Show that:

$$\frac{d\sigma}{d\Omega} = \frac{1}{4\gamma^2 \sin^4(\phi/2)}$$

As a point of curiosity, this quantity is called the *differential cross section*.

Solution

a) The diagram of the situation is the following:

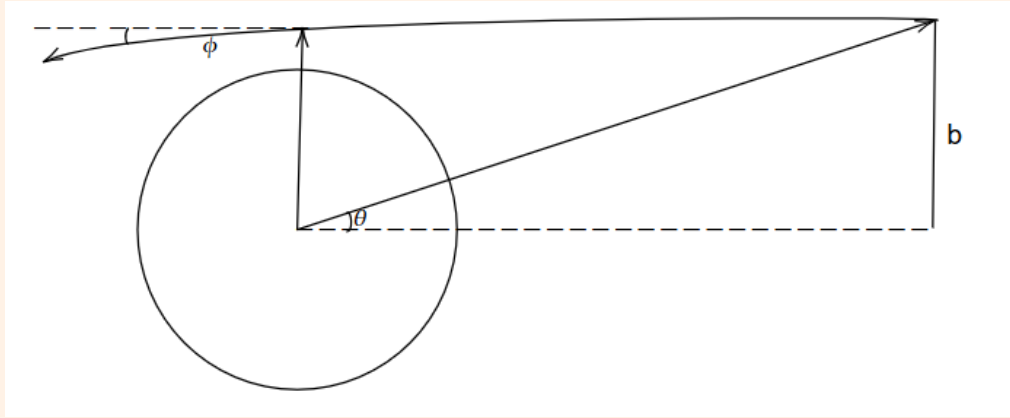


Figure 3: Diagram of the hyperbolic orbit

To find the deflection angle ϕ , we will use the impulse theorem, considering the horizontal and vertical components of the attractive force:

$$F_x = -\frac{GMm}{r^2} \cos \theta, \quad F_y = -\frac{GMm}{r^2} \sin \theta$$

The impulse theorem tells us that:

$$\int F_i dt = \Delta p$$

where p_i is the momentum of the particle in the direction i .

On the x -axis, we have:

$$-\int \frac{GMm}{r^2} \cos \theta dt = \Delta p_x$$

Note that integrating with respect to time makes things inconvenient, because finding how θ and r vary in time is complicated. For this reason, we turn to the definition of angular momentum. Using $L = mr^2 d\theta/dt$, we have:

$$dt = \frac{mr^2}{L} d\theta$$

Substituting this value into our integral:

$$-\int \frac{GMm^2}{L} \cos \theta d\theta = \Delta p_x$$

Notice that our integral goes from $\theta = 0$, the situation at infinity, to $\theta = \pi - \phi$, the situation at infinity after the deflection. Thus:

$$-\frac{GMm^2}{L} \int_0^{\pi-\phi} \cos \theta d\theta = -\frac{GMm^2}{L} \sin \theta \Big|_0^{\pi-\phi} = -\frac{GMm^2}{L} \sin(\pi - \phi) = \Delta p_x$$

Analogously for y :

$$\begin{aligned} & -\frac{GMm^2}{L} \int_0^{\pi-\phi} \sin \theta d\theta = \Delta p_y \\ & = \frac{GMm^2}{L} \cos \theta \Big|_0^{\pi-\phi} \equiv -\frac{GMm^2}{L} (1 - \cos(\pi - \phi)) = \Delta p_y \end{aligned}$$

Also note that $\Delta p_x = m(v_{f,x} - v_0)$, $\Delta p_y = m(v_{f,y} - 0)$. We can also find a relation between ϕ and the velocities and, by pure geometry, we obtain:

$$\tan \phi = -\frac{v_{f,y}}{v_{f,x}}$$

Since there are only central forces, L is constant, and from the initial situation it equals $L = mbv_0$. Now, turning to the calculation:

$$-\frac{GM}{bv_0} \sin(\pi - \phi) = v_{f,x} - v_0, \quad -\frac{GM}{bv_0} (1 - \cos(\pi - \phi)) = v_y$$

With these two equations, we obtain:

$$\frac{v_{f,y}}{v_{f,x}} = -\frac{GM(1 - \cos(\pi - \phi))}{bv_0^2 - GM \sin(\pi - \phi)}$$

That is:

$$\tan \phi = \frac{GM(1 - \cos(\pi - \phi))}{bv_0^2 - GM \sin(\pi - \phi)}$$

Solving for ϕ , we obtain the expected result. Another way to carry out the same question is to think purely about the geometry of the situation:

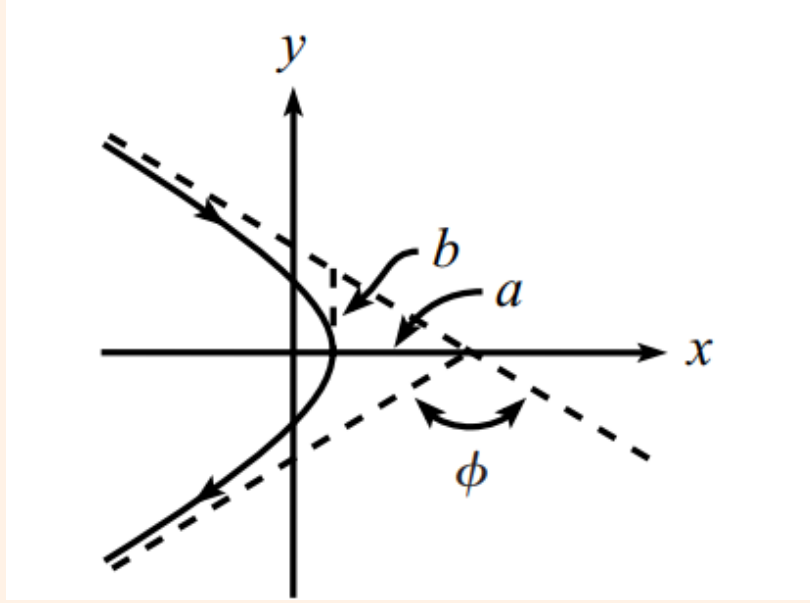


Figure 4: Source: Morin 7.7

Here, it is evident that:

$$\phi = \pi - 2 \tan^{-1} \left(\frac{b}{a} \right)$$

where b is the impact parameter and a the semiaxis of the hyperbola. From the theory of conics, we have:

$$\frac{b}{a} = \sqrt{e^2 - 1} = \sqrt{\frac{2EL^2}{m(GMm)^2}} = \frac{v_0^2 b}{GM} \equiv \gamma b$$

In both approaches, we arrive at the same result:

$$\boxed{\phi = \pi - \tan^{-1}(\gamma b)}$$

as desired.

- b)** Consider a ring of thickness db and radius b . Now consider a very large sphere, centered at M . Any particle that passes through the cross section of the ring of radius b will hit this sphere making an angle ϕ with the x -axis, with an angular separation $d\phi$. Using $d \cot x / dx = -1 / \sin^2 x$, we have:

$$\left| \frac{db}{d\phi} \right| = \frac{1}{2\gamma \sin^2(\phi/2)}$$

The cross-sectional area is given by $d\sigma = 2\pi b db$ and the solid angle on a sphere is given by $d\Omega = 2\pi \sin \phi d\phi$. Therefore:

$$\frac{d\sigma}{d\Omega} = \frac{2\pi b}{2\pi \sin \phi} \frac{db}{d\phi} = \left(\frac{(1/\gamma) \cot(\phi/2)}{2 \sin(\phi/2) \cos(\phi/2)} \right) \left(\frac{1}{2\gamma \sin^2(\phi/2)} \right)$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{1}{4\gamma^2 \sin^4(\phi/2)}}$$

Idea 1

In certain questions, the use of vectors is appropriate. One example is the question below. To make things easier, when finding the inclination of the orbit, look at the following figure:

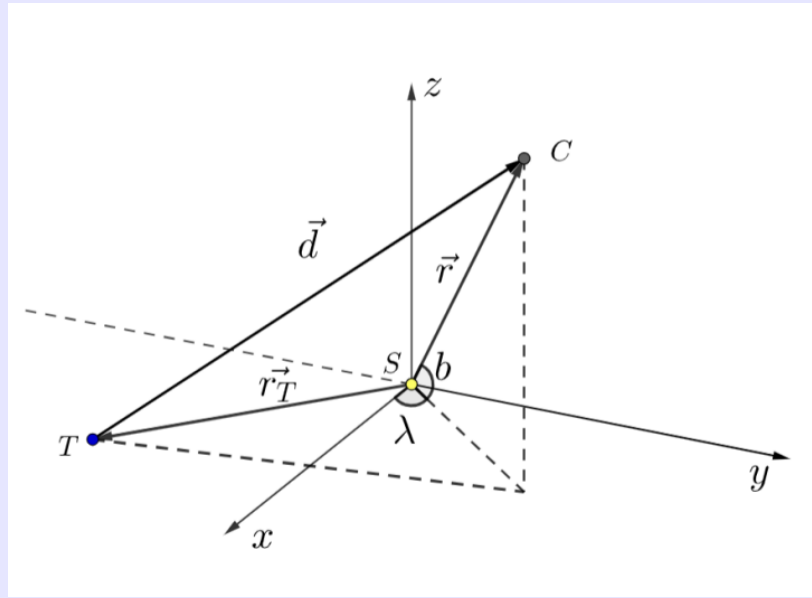


Figure 5: Vectors

Here, C represents a body, T the Earth, and S the Sun. The angles λ and b denote, respectively, the ecliptic longitude and the ecliptic latitude. $\vec{r}$ is the vector between the Sun and the body, $\vec{r}_T$ the vector between the Sun and the Earth, and $\vec{d}$ the vector between the Earth and the body. From the figure,

$$\vec{r} = r(\cos b \cos \lambda \hat{x} + \cos b \sin \lambda \hat{y} + \sin b \hat{z})$$

$$\vec{r}_T = r_T(\cos(\omega_{\oplus} t) \hat{x} + \sin(\omega_{\oplus} t) \hat{y})$$

Thus, $\vec{d} = \vec{r} - \vec{r}_T$:

$$\vec{d} = (r \cos b \cos \lambda - r_T \cos(\omega_{\oplus} t)) \hat{x} + (r \cos b \sin \lambda - r_T \sin(\omega_{\oplus} t)) \hat{y} + r \sin b \hat{z}$$

Calculating the magnitude of the vector $\vec{d}$, we obtain:

$$d = |\vec{d}| = \sqrt{r_T^2 + r^2 - 2r_T r \cos b \cos(\lambda - \omega_{\oplus} t)}$$

Therefore:

$$\cos b \cos(\lambda - \omega_{\oplus} t) = \frac{r_T^2 + r^2 - d^2}{2r_T r}$$

- [5] **Problem 12.** (Problem set 1 - Vinhedo 2024) CR4b-2023 is a probe on a heliocentric orbit that was built and launched in 2023 by the young prodigy Caranguejo. The goal of the probe was to study solar storms and collect data for Caranguejo to analyze at his observatory in Espírito Santo. CR4b-2023, however, during one of its expeditions toward the Sun, was hit by a solar storm and suffered serious damage. Because of this, the probe became disoriented and thus had all of its orbital parameters changed, so that Caranguejo no longer knew its location.

Aiming to track the position of CR4b-2023 again, Caranguejo used his observatory to collect the values of the angular separations $\Delta\phi$ between the probe and the Sun and the values of the angular diameter θ_S of the probe, all as a function of time t . The table obtained by Caranguejo is shown below.

$\Delta\phi$ (Degrees)	θ_S (mas)	t (Days)
0.000	99.27	0.00
4.889	103.67	1.79
17.354	104.72	7.43
29.015	93.06	20.62
27.793	81.18	25.41
13.460	79.77	45.98
11.539	77.97	51.61
2.466	76.44	54.53
2.656	74.84	55.31
7.177	73.90	57.19
10.800	70.63	58.43
22.464	72.92	91.67
13.823	80.50	130.48

Table 1: Values measured by Caranguejo

Assume that, at the initial instant $t_0 = 0$ when the probe is hit by the solar storm, the Sun was exactly at the point of Libra and the motion of the probe was ascending relative to the ecliptic plane. Also assume that the radius of the probe, supposed spherical, is given by $R = 30$ km. For this question, an error analysis is not required (although it is important to try to use methods aimed at reducing statistical errors). Based on what was presented:

- (a) Calculate the orbital parameters of the orbit of CR4b-2023, that is, its semimajor axis a , eccentricity e , inclination i , longitude of the ascending node Ω , and argument of perihelion ω . As in all data-analysis questions, you must provide clearly labeled data tables, clearly labeled graphs, and enough formula derivations to make clear what you measured and how you are deriving your results in order to reduce statistical errors.

- (b) Taking (x', y', z') as the coordinates of CR4b-2023 in a right-handed Cartesian system in which the $x'y'$ plane lies in the plane of the probe's orbit, the Sun is at the origin, and the x' axis points toward the vernal point, write, in a table, the values of at least 5 distinct points (x', y', z') . Based on the data found, sketch, on a graph, the orbit of CR4b-2023, indicating the coordinates of its center.
- (c) Now considering a new right-handed coordinate system (x, y, z) , in which the Sun is at the origin, the xy plane represents the ecliptic plane, and the x axis points toward the vernal point, write the coordinates of the center of the orbit of CR4b-2023. Finally, calculate the distance d_{TC} between the center of Earth's orbit and the center of the probe's orbit.

Solution

- a) The idea of this part of the question is to find a relation between the days and the distance from the probe to the Sun. Using basic trigonometry, we can say that the distance from the probe to the Earth is given by:

$$d_{S-\oplus} = R / \tan^{-1} \left(\frac{\theta_S}{2} \right)$$

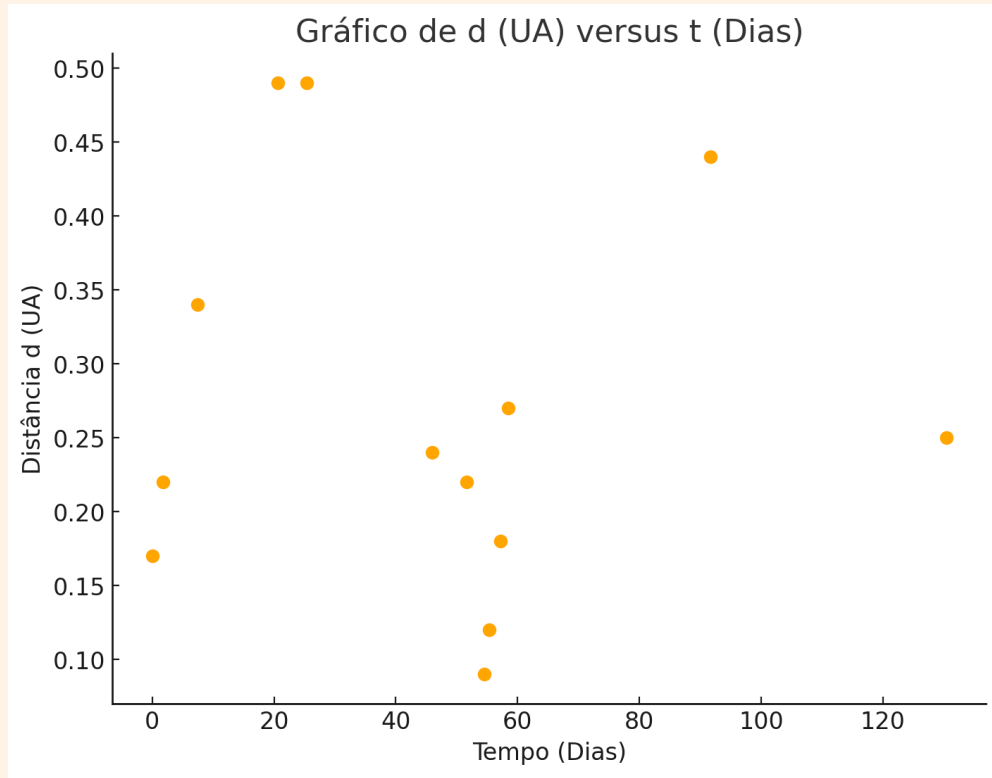
where R is the radius of the probe. Here, we can use the law of cosines to relate this distance to the distance from the probe to the Sun. Considering the Earth's orbit to be circular with radius $d_{\odot-\oplus}$ and defining d as the distance between the probe and the Sun, we have:

$$d^2 = d_{\odot-\oplus}^2 + d_{S-\oplus}^2 - 2d_{\odot-\oplus}d_{S-\oplus} \cos \Delta\phi$$

With this, we can build the following table, with the values of d in AU and the values of t in days.

t (Days)	d (UA)
0.00	0.17
1.79	0.22
7.43	0.34
20.62	0.50
25.41	0.49
45.98	0.24
51.61	0.22
54.53	0.09
55.31	0.12
57.19	0.18
58.43	0.27
91.67	0.44
130.48	0.25

Plotting a graph of t versus d , we have:



Now let us get to work and find the requested parameters: semimajor axis, a , eccentricity, e , inclination, i , longitude of the ascending node Ω , and argument of perihelion, ω . The first two can be obtained directly from the data we have analyzed so far.

Analyzing the data, we can see that the smallest distance between the probe and the Sun is given by $a_p = 0.09$ AU and the largest distance is given by $a_a = 0.49$ AU. These distances must correspond to perihelion and aphelion, respectively. Using the definition:

$$a_p = a(1 - e), \quad a_a = a(1 + e) \rightarrow \frac{a_p}{a_a} = \frac{1 - e}{1 + e} \approx 0.18$$

Solving, we obtain $e \approx 0.67$.

Using $a = \frac{a_p + a_a}{2}$, we obtain $a \approx 0.30$ AU.

To find the angular factors, we have to think a bit more. Starting with Ω , note that at the initial instant, at $t = 0$, the angular separation between the Sun and the probe is $\Delta\phi = 0$. The statement tells us that at this moment the Sun is at the point of Libra, so the probe is also at the point of Libra and, since its motion was ascending, we have $\Omega = 0^\circ$.

To find ω and i , let us look at the following spherical triangle:

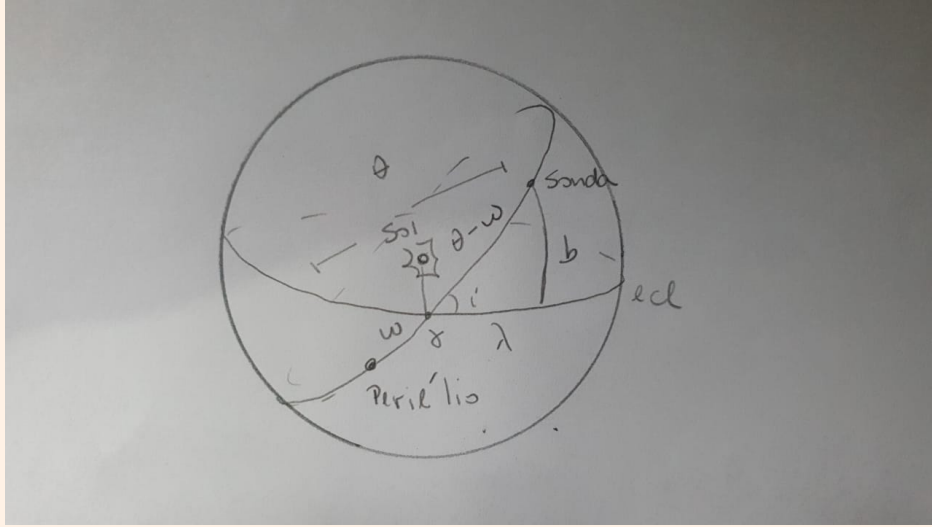


Figure 6: Diagram

From the polar form of the ellipse, $r = \frac{a(1-e^2)}{1+e\cos\theta}$, where θ is the true anomaly of the orbit, we can find ω . Notice that at the initial instant, $t = 0$, the probe is exactly at the point of Libra, so $\theta = 2\pi - \omega$ as in the diagram.

Thus, we have:

$$d(0) = \frac{a(1-e^2)}{1+e\cos(2\pi-\omega)}$$

Solving for ω :

$$\omega = 2\pi - \cos^{-1} \left(\frac{a(1-e^2)}{d(0)e} - \frac{1}{e} \right) \approx 267.6^\circ \approx 270^\circ$$

substituting the values found.

To find the inclination, we will use the law of cosines:

$$\cos(\theta - \omega) = \cos \lambda \cos b + \sin \lambda \sin b \cos(\pi/2) = \cos \lambda \cos b$$

Note that $\cos(\theta - \omega) \approx \cos(\theta - 270^\circ) = \sin \theta$.

Rewriting,

$$\cos b = \frac{\sin \theta}{\cos \lambda}$$

Using the relation obtained in the previous idea,

$$\cos b = \frac{r_T^2 + r^2 - d^2}{2r_T r \cos(\lambda - \omega_{\oplus} t)}$$

Equating these expressions and solving for λ ,

$$\lambda = \tan^{-1} \left(\frac{r_T^2 + r^2 - d^2}{2r_T r \sin \theta \sin(\omega_{\oplus} t)} - \cot(\omega_{\oplus} t) \right)$$

where:

$$\theta = \arccos \left(\frac{a(1 - e^2)}{er} - \frac{1}{e} \right)$$

With this, we can make the following table, with the values of b and λ .

λ (Degrees)	b (Degrees)
0.000	0.00
17.495	9.847
45.905	22.521
78.492	29.499
101.508	29.499
134.095	22.521
163.504	9.847
180.000	0.000
197.495	-9.847
216.005	-18.747
270	-30
333.435	-14.478
17.495	9.847
78.492	29.499
143.994	18.747

Returning to the spherical triangle and using the four-parts formula, we have:

$$\cos \lambda \cos 90^\circ = \sin \lambda \cot b - \sin 90^\circ \cot i$$

Simplifying:

$$\tan b = \tan i \sin \lambda$$

Note that this resembles a function of the type $y = A + Bx$. Performing a linear regression on a calculator, we find $B \approx 0.578$. Since $\tan i = B$, we arrive at:

$$i \approx 30^\circ$$

b) Since the Sun is at the origin, the values of x' and y' can be calculated by:

$$x' = r \sin \theta$$

and

$$y' = -r \cos \theta$$

Writing five different values of x' and y' in a table, we obtain:

Table 7: Values of x' and y'

x' AU	y' AU
0.167	0.000
0.084	0.478
-0.219	0.261
-0.203	0.074
0.000	-0.100

Notice that any values of x' and y' , provided they are correct, may be used. Furthermore, it is important to increase the spacing between the values of x' and y' in order to reduce the errors associated with sketching the orbit. Based on this, we can plot the graph:

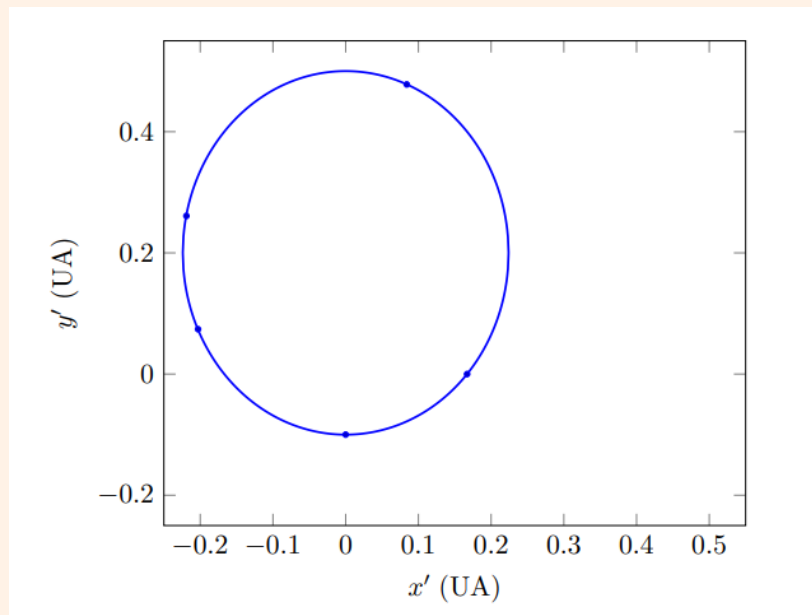


Figure 7: Graph of the orbit

Observing the graph, we see that the coordinates (x', y', z') of the center of the ellipse are $(0; 0.2; 0)$.

c) Here, we will use a rotation matrix about the x -axis, given by:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

For $\alpha = i = 30^\circ$:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & \sin 30^\circ \\ 0 & -\sin 30^\circ & \cos 30^\circ \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

Then:

$$\begin{aligned} x &= x' \\ y &= y' \cos 30^\circ + z' \sin 30^\circ \\ z &= -y' \sin 30^\circ + z' \cos 30^\circ \end{aligned}$$

Substituting the numerical values, we obtain that the coordinates (x, y, z) of the center of CR4b-2023 are given by:

$$(0.000, 0.173, -0.100).$$

Finally, we obtain that the distance between the center of Earth's orbit and the center of the probe's orbit is given by:

$$d = \sqrt{|x^2 + y^2 + z^2|}$$

that is:

$$d = 0.200 \text{ UA}$$

- [5] **Problem 13.** (Iran Problem Set) A particle of mass m is orbiting a massive object of mass M . Show that the impulse required to rotate the orbit by an angle η about one of the foci is given by:

$$\Delta v = \frac{2\mu e}{h} \sin\left(\frac{\eta}{2}\right),$$

where:

- h is the angular momentum per unit mass;
- $\mu = GM$, where G is the gravitational constant.

Solution

Drawing the orbit and the rotated orbit:

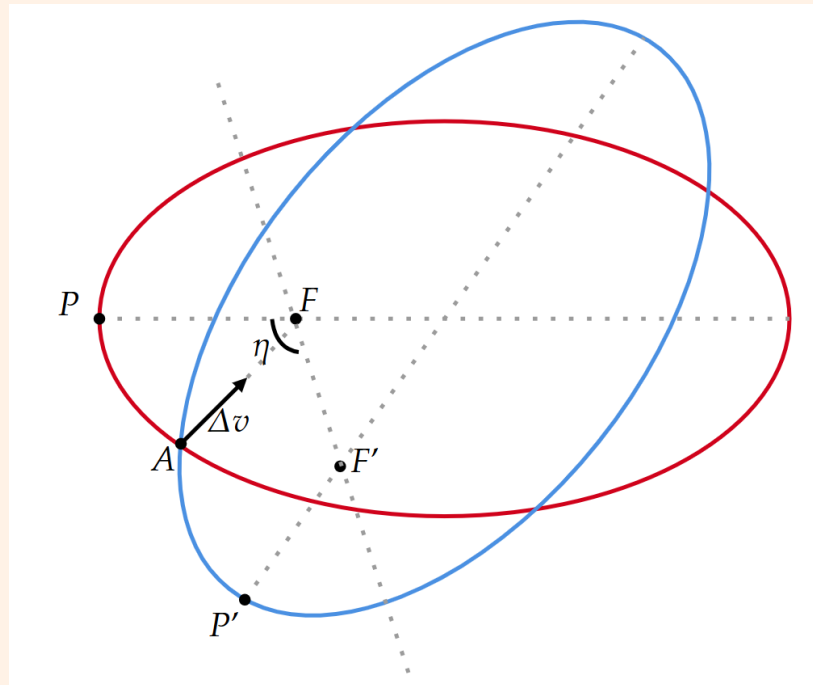


Figure 8: Diagram of the rotated orbit.

Notice that the impulse can be applied only at the intersection points of the two orbits and must be radial, since the angular momentum must remain constant in order to keep the orbit identical. Let us define X as a point on the orbit before the impulse and X' as a point on the orbit after the rotation.

Let us set the following definitions: F is the focus, P is the perihelion of the orbit, and A is the intersection point at which the impulse is applied.

By symmetry, note that the angles AFP and AFF' are equal and therefore have the value $\eta/2$.

Since the two orbits have the same angular momentum, the tangential velocity at the point A must be the same for both orbits. Furthermore, we can see that the point A has true anomaly $\theta = \eta/2$, which guarantees that the speed is also the same for both orbits.

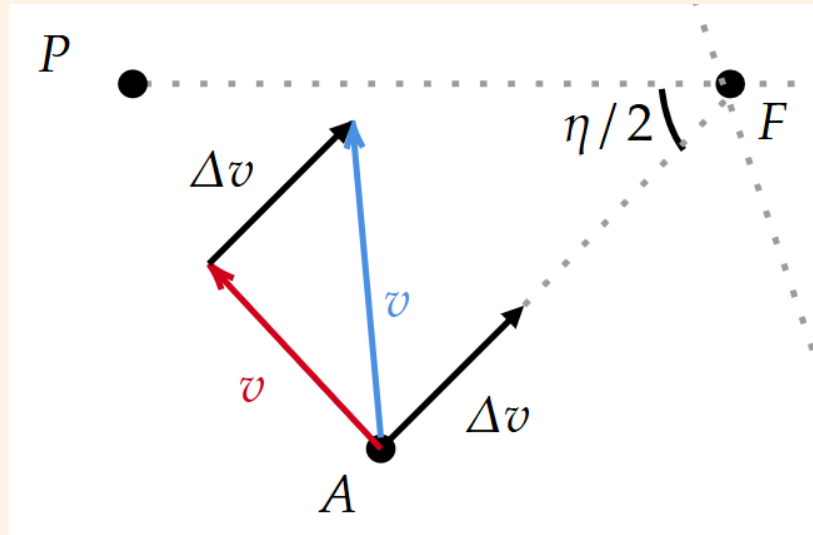


Figure 9: Velocity vectors.

The only way for this to be possible is if the radial impulse is given by $\Delta v = 2v_r$, where v_r is the radial velocity immediately before the impulse.

Of course $v_r^2 = v^2 - v_t^2$, where v_t is the tangential velocity. Starting from basic definitions:

$$h^2 = \mu a(1 - e^2), \quad r = \frac{a(1 - e^2)}{1 - e \cos \theta}, \quad v_t = \frac{h}{r}, \quad v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right).$$

Working with these equations, we arrive at:

$$v^2 = \frac{\mu^2}{h^2}(1 + 2e \cos \theta + e^2), \quad v_t^2 = \frac{\mu^2}{h^2}(1 + e \cos \theta)^2.$$

Finally, substituting into $v_r^2 = v^2 - v_t^2$, we obtain:

$$v_r^2 = \frac{\mu^2}{h^2}(e^2 - e^2 \cos^2 \theta) \equiv \frac{\mu^2 e^2}{h^2} \sin^2 \theta.$$

Substituting $\Delta v = 2v_r$ and $\theta = \eta/2$, we arrive at the desired result:

$$\Delta v = \frac{2\mu e}{h} \sin \left(\frac{\eta}{2} \right) \blacksquare$$

- [3] **Problem 14.** In this question, we will study a simplified model of the effect that confirmed Einstein's theory of general relativity: gravitational lensing. The presence of a massive body curves spacetime, so that stars can serve as lenses in space. Some telescopes, such as the JWST, use this effect to photograph very distant galaxy clusters. One of the photographs taken by the JWST of a gravitational lens is shown below:



Figure 10: Source: National Geographic

A simplified diagram of gravitational lensing can be seen below. This “circle” is known as the Einstein ring.

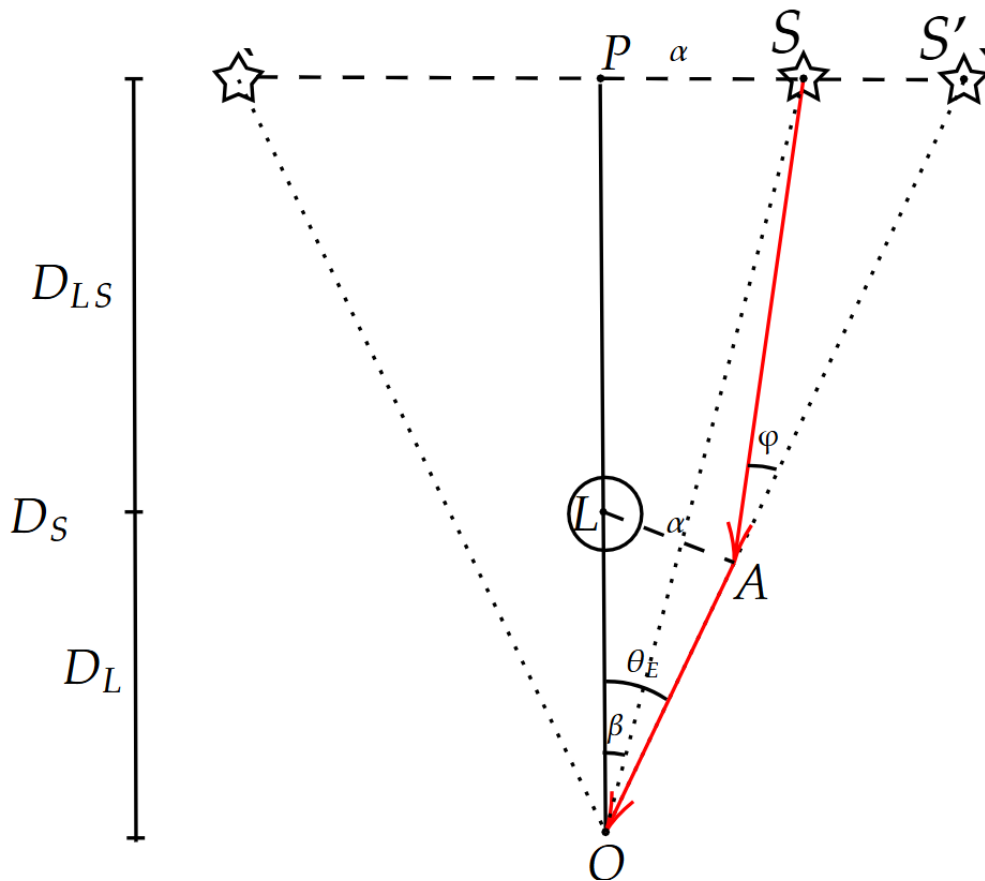


Figure 11: Diagram of the gravitational lens

In the diagram, the point O represents the observer, L the massive body that acts as a lens, S the true position of the observed object, and S' its apparent position. α is the impact parameter, β the angular separation between the observed body and the lens, θ_E the angular radius of the Einstein ring, and φ the angular deflection caused by a massive body. The distance $\overline{OL}$ equals D_L and the distance $\overline{OP}$ equals D_S , so that $D_S - D_L = D_{LS}$.

All the angles are very small, so that $\sin x \approx \tan x \approx x$. Furthermore, because they lie at infinity, the lines $\overline{OP}$, $\overline{OS}$, and $\overline{OS'}$ may be approximated as parallel.

It is a known result of the theory of general relativity that the deflection of light caused by a massive body is given by:

$$\varphi = \frac{4GM}{\alpha c^2}$$

- In the limit $\lim_{\alpha \rightarrow 0}$, find an expression for θ_E in terms of the other distances and angles provided.
- The JWST telescope obtains images in the infrared, of wavelength λ_{IV} . How large must its diameter be so that it can resolve an Einstein ring?

Solution

a) Note that, when $\alpha \rightarrow 0$, $\overline{PL} = \overline{AS} = D_{LS}$. Using $\tan \theta = \sin \theta = \theta$, we have:

$$\begin{aligned} (1) \overline{PS} &= D_S \beta = \alpha, & (2) \overline{SS'} &= D_{SL} \varphi, & (3) \overline{PS'} &= D_S \theta_E, \\ (4) \overline{OL} &= D_L = \overline{LA} / \theta_E = \alpha \theta_E, & (5) \overline{LA} &= \overline{PS}, & (6) \overline{PS'} &= \overline{PS} + \overline{SS'} \end{aligned}$$

Substituting (1), (2), (3) into (6),

$$D_S \theta_E = D_S \beta + (D_S - D_L) \varphi$$

$$\theta_E - \beta = \frac{D_S - D_L}{D_S} \varphi$$

Substituting the value of φ :

$$\theta_E - \beta = \frac{D_S - D_L}{D_S} \frac{4GM}{\alpha c^2}$$

Using (4), we have $\alpha = D_L \theta_E$, and substituting into our equation:

$$\theta_E - \beta = \frac{D_S - D_L}{D_S} \frac{4GM}{D_L \theta_E c^2}$$

$$\theta_E^2 - \beta \theta_E = \frac{D_S - D_L}{D_S} \frac{4GM}{c^2}$$

The limit $\alpha \rightarrow 0$ implies that $\beta \rightarrow 0$, so:

$$\theta_E = \sqrt{\frac{D_S - D_L}{D_S} \frac{4GM}{c^2}}$$

b) Using the Rayleigh criterion $\theta = R \frac{\lambda}{D}$, where $R \approx 1.22$. Isolating D and substituting $\theta = \theta_E$:

$$D = \sqrt{\frac{c^2 R^2 \lambda_{IV}^2}{4GM} \frac{D_S}{D_S - D_L}}$$

[4] **Problem 15.** In this exercise, we will study properties of the eccentricity of orbits and understand how it relates to energy and angular momentum. Denoting $\mu = GM$ and $h = L/m$, do what is asked in the following items.

Part I: The eccentricity vector

a) Write an expression for Newton's second law in vector form for a body of mass m orbiting a body of mass M at a distance r .

- b) Take the cross product of your expression with $\mathbf{h}$ and prove that:

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \frac{\mu}{r} \mathbf{v} - \frac{\mu \dot{r}}{r^2} \mathbf{r}$$

where $\dot{\mathbf{r}}$ represents the radial velocity and $\mathbf{v}$ the velocity.

- c) We can write the right-hand side of the previous equation as:

$$\frac{d}{dt}(A\mathbf{r})$$

Find the value of A .

- d) Integrate both sides of the equality you found and take the scalar product of both sides with $\mathbf{r}$. After that, solve for r . Your result should be very similar to what is predicted by geometry alone:

$$r = \frac{p}{1 + e \cos \theta}$$

Find the values of p and e . **Do not forget the integration constant. Your answers may depend on it.**

- e) In the previous item, e is the eccentricity of the orbit. Returning to the equation initially obtained in item c, after the integration, find a way to express $\mathbf{e}$ as a vector of magnitude e that points directly toward the orbited body.
- f) Prove that the relation found in the previous item can be written as:

$$\mu \mathbf{e} = (v^2 - \frac{\mu}{r}) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v}$$

Where does this vector point?

Part II: Determining orbital elements from the eccentricity vector

Let us define our coordinate system as follows. Consider a right-handed system, oriented so that the xy plane coincides with the equatorial plane and $\mathbf{z}$ points toward the north pole. Define the nodal vector as $\mathbf{n} = \mathbf{z} \times \mathbf{h}$.

- a) Find an expression for $\mathbf{h}$ in terms of the components of the vector $\mathbf{r}$ and of the vector $\mathbf{v}$. After that, find the vector $\mathbf{n}$ in terms of the components of $\mathbf{h}$.
- b) Draw a celestial sphere and represent on it: the equatorial plane and an arbitrary orbit (with inclination different from 0). On it, mark the following angles: i , the orbital inclination, Ω the longitude of the ascending node, ω the argument of periastron, and θ , the true anomaly.
- c) Find expressions for all of these angles in terms of $\mathbf{e}$, $\mathbf{h}$, $\mathbf{n}$, $\mathbf{r}$, and any of the vectors defined in our xyz coordinate system.

Solution**Part I**

a) Using the vector form of the gravitational force, we have:

$$-\frac{GMm}{r^3}\mathbf{r} = m\ddot{\mathbf{r}} \rightarrow \boxed{\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}}$$

b) Taking the cross product of this expression with $\mathbf{h}$, we have:

$$\ddot{\mathbf{r}} \times \mathbf{h} = -\frac{\mu}{r^3}(\mathbf{r} \times \mathbf{h}) = \frac{\mu}{r^3}(\mathbf{h} \times \mathbf{r})$$

Since $\mathbf{h}$ is constant, the left-hand side of the equation can be written as $d(\dot{\mathbf{r}} \times \mathbf{h})/dt$. For the right-hand side, let us write $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ and use the BAC-CAB rule, which states that $(A \times B) \times C = B(A \cdot C) - C(A \cdot B)$. Thus:

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \frac{\mu}{r^3}((\mathbf{r} \times \mathbf{v}) \times \mathbf{r}) = \frac{\mu}{r^3}(r^2\mathbf{v} - \mathbf{r}(\mathbf{r} \cdot \mathbf{v}))$$

Note that $\mathbf{r} \cdot \mathbf{v}$ is the product of the components of $\mathbf{r}$ and $\mathbf{v}$ that lie in the same direction. Thus, $\mathbf{r} \cdot \mathbf{v} = r\dot{r}$. Substituting into the previous expression:

$$\boxed{\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \frac{\mu}{r}\mathbf{v} - \frac{\mu\dot{r}}{r^2}\mathbf{r}}$$

c) Expanding the expression given in the statement:

$$\frac{d}{dt}(A\mathbf{r}) = \mathbf{r}\frac{d}{dt}A + A\mathbf{v}$$

Comparing the expressions, it is easy to see that $\boxed{A = \mu/r}$.

d) Using the information obtained in the previous items, we have:

$$\frac{d}{dt}(\dot{\mathbf{r}} \times \mathbf{h}) = \mu\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right)$$

Multiplying both sides by dt and integrating, we have:

$$\dot{\mathbf{r}} \times \mathbf{h} = \frac{\mu}{r}\mathbf{r} + \mathbf{B}$$

where $\mathbf{B}$ is a constant vector of integration. Now, taking the scalar product with $\mathbf{r}$:

$$\mathbf{r} \cdot (\dot{\mathbf{r}} \times \mathbf{h}) = \mu r + \mathbf{r} \cdot \mathbf{B}$$

Using $a \cdot (b \times c) = (a \times b) \cdot c$ and defining ν as the angle between $\mathbf{r}$ and $\mathbf{B}$, we have:

$$h^2 = \mu r + rB \cos \nu$$

Solving for r , we arrive at:

$$r = \frac{h^2/\mu}{1 + (B/\mu) \cos \nu}$$

Thus:

$$\boxed{p = \frac{h^2}{\mu}, \quad e = \frac{B}{\mu}}$$

e) Using $\mathbf{e} = \mu \mathbf{B}$ and returning to item d), we have:

$$\dot{\mathbf{r}} \times \mathbf{h} = \frac{\mu}{r} \mathbf{r} + \mu \mathbf{e}$$

Solving for $\mathbf{e}$ and noting that $\dot{\mathbf{r}} \times \mathbf{h} = \mathbf{v} \times \mathbf{h}$, we have:

$$\boxed{\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}}$$

f) Writing $\mathbf{h} = \mathbf{r} \times \mathbf{v}$, we have:

$$\mathbf{e} = \frac{\mathbf{v} \times (\mathbf{r} \times \mathbf{v})}{\mu} - \frac{\mathbf{r}}{r}$$

Multiplying both sides of the expression by μ , and using BAC-CAB, we have:

$$\mu \mathbf{e} = v^2 \mathbf{r} - \mathbf{v}(\mathbf{v} \cdot \mathbf{r}) - \frac{\mu}{r} \mathbf{r}$$

Rearranging the terms:

$$\boxed{\mu \mathbf{e} = (v^2 - \frac{\mu}{r}) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v}}$$

Thus, this vector points toward periastron.

Part II: Determining orbital elements from the eccentricity vector

a) The vector $\mathbf{h}$ is defined by $\mathbf{h} = \mathbf{r} \times \mathbf{v}$. By definition,

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ r_x & r_y & r_z \\ v_x & v_y & v_z \end{vmatrix} = (r_y v_z - r_z v_y) \hat{x} + (r_z v_x - r_x v_z) \hat{y} + (r_x v_y - r_y v_z) \hat{z}$$

Similarly, $\mathbf{n} = \mathbf{z} \times \mathbf{h}$

$$\mathbf{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ h_x & h_y & h_z \end{vmatrix} = -h_y \hat{x} + h_x \hat{y}$$

b) The drawing should resemble the following figure:

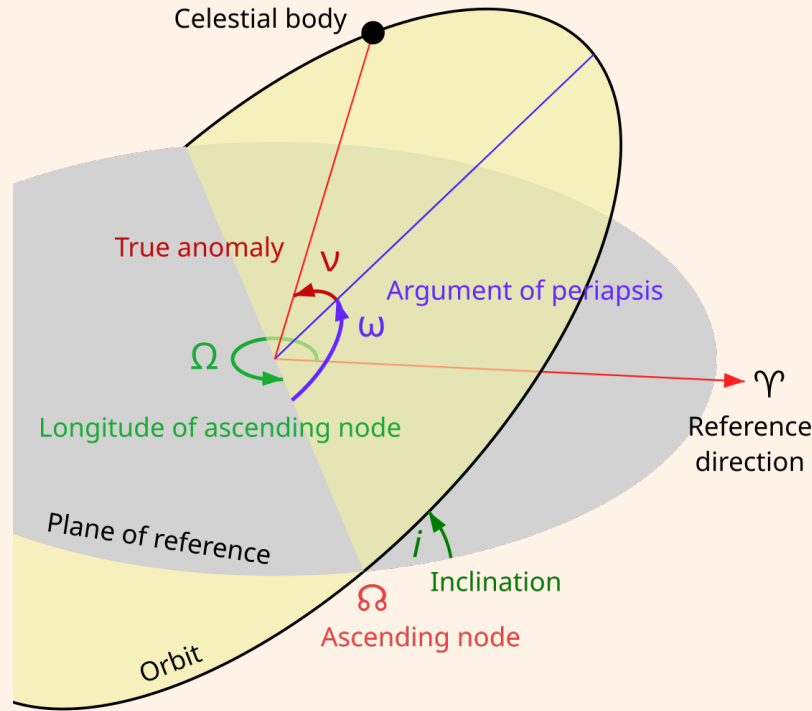


Figure 12: Source: Wikipedia

c) To orient ourselves, let us create a coordinate system in which the north pole corresponds to the z -axis and the x -axis points in the direction of the vernal point. Starting with the inclination, it is the angle between the north pole and the plane perpendicular to the orbit. Thus

$$\mathbf{z} \cdot \mathbf{h} = \cos i |\mathbf{z}| |\mathbf{h}| \rightarrow i = \cos^{-1} \left(\frac{\mathbf{z} \cdot \mathbf{h}}{|\mathbf{z}| |\mathbf{h}|} \right)$$

Using similar arguments, we can obtain

$$\Omega = \cos^{-1} \left(\frac{\mathbf{x} \cdot \mathbf{n}}{|\mathbf{x}| |\mathbf{n}|} \right)$$

$$\omega = \cos^{-1} \left(\frac{\mathbf{n} \cdot \mathbf{e}}{|\mathbf{n}| |\mathbf{e}|} \right)$$

$$\theta = \cos^{-1} \left(\frac{\mathbf{e} \cdot \mathbf{r}}{|\mathbf{e}| |\mathbf{r}|} \right)$$

[5] **Problem 16.** ★ ★ ★ (Adapted from IPhO 2018) One of the most interesting effects of general relativity is the emission of gravitational waves (GWs) by close binaries. One consequence of this is the loss of energy of the system. In this question, we will study this effect in depth.

a) Consider a system formed by stars 1 and 2, with masses M_i and at distances r_i from the

center of mass. Using Newton's second law, one can show that:

$$\mathbf{a}_1 = -\alpha \frac{\mathbf{r}_1}{r_1^n}$$

where $\mathbf{a}_1$ is the acceleration vector of star 1. Find the value of $\alpha = \alpha(G, M_1, M_2)$ (read as α in terms of G, M_1, M_2) and of n .

b) The total energy of the binary system can be expressed as:

$$E = A(\omega, \mu, L) - \frac{GM\mu}{L}$$

where μ is the reduced mass of the system, $M = M_1 + M_2$, and $L = r_1 + r_2$. Find the value of A .

c) The previous expression can be simplified to:

$$E = \frac{\beta GM\mu}{L}$$

where β is a dimensionless constant. Find its value.

The correct theory of gravity, general relativity, was formulated by Einstein in 1915 and predicts that gravity propagates at the speed of light. The messengers that carry information about the interaction are called gravitational waves (GWs). GWs are emitted whenever masses are accelerated, causing the system of masses to lose energy.

Consider a system of two point particles, isolated from the rest of the Universe. Einstein proved that, for sufficiently small speeds, the emitted waves: 1) have a frequency twice the orbital frequency; 2) can be characterized by a luminosity, that is, an emitted power $\mathcal{P}$, which is dominated by the formula:

$$\mathcal{P} = \frac{G}{5c^5} \sum_{i=1}^3 \sum_{j=1}^3 \left(\frac{d^3 Q_{ij}}{dt^3} \right) \left(\frac{d^3 Q_{ij}}{dt^3} \right).$$

Here, c is the speed of light $c \simeq 3 \times 10^8$ m/s. For a system of two point particles orbiting in the $x - y$ plane, Q_{ij} is the following table (i, j indicate the row/column number):

The components of Q_{ij} are given by:

$$Q_{11} = \sum_{A=1}^2 \frac{M_A}{3} (2x_A^2 - y_A^2),$$

$$Q_{22} = \sum_{A=1}^2 \frac{M_A}{3} (2y_A^2 - x_A^2),$$

$$Q_{33} = - \sum_{A=1}^2 \frac{M_A}{3} (x_A^2 + y_A^2),$$

$$Q_{12} = Q_{21} = \sum_{A=1}^2 M_A x_A y_A.$$

And $Q_{i,j} = 0$ for all other possibilities. Here, x_A, y_A are the positions of star A , with $A = 1, 2$, in a Cartesian plane with the center of mass at the origin.

- d) Write the coordinates (x_1, y_1) and (x_2, y_2) in terms of r_1, r_2, ω , and t , with t the time elapsed since some specific point of the orbit.
- e) To calculate the dissipated power, the formula contains a quadruple sum. Solving this expression by brute force is quite complicated and time-consuming. Fortunately, there is a nicer and more elegant way to solve it. To do so, begin by writing $Q_{i,j}$ as a 3x3 matrix, of the following form:

$$Q = A \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{23} & a_{33} \end{pmatrix}$$

Find the coefficient $A = A(\mu, L)$ and complete the matrix Q .

- f) From the previous item, you should obtain:

$$Q_{ii} = A(b_i + j_i \cos(kt)), \quad Q_{ij}^{i \neq j} = A(p_{ij} \sin(kt))$$

Find the values of b_i, j_i, p_{ij} , and k .

- g) You are now able to solve for $\mathcal{P}$ more smoothly. The expression can be simplified to:

$$\mathcal{P} = \xi \frac{G}{c^5} \mu^2 L^4 \omega^6$$

Find the numerical value of ξ .

HINT: The double sum:

$$\sum_{i=1}^N \sum_{j=1}^N (A_{ij})(A_{ij})$$

where A is an $N \times N$ matrix, can be simplified to:

$$\sum_{i=1}^N \sum_{j=1}^N A_{ij}^2$$

which represents the sum of the squares of all the elements of the matrix A .

- h) If there were no emission of GWs, the system would remain in equilibrium indefinitely, but because of that emission the energy of the system is not conserved, so that there is a change

in the angular velocity of the system, ω . The formula for the time variation of ω has the following form:

$$\left(\frac{d\omega}{dt}\right)^3 = (3\xi)^3 \frac{\omega^{11}}{c^{15}} (GM_C)^5$$

where M_c is the so-called *chirp* mass and $M_c = M_c(M, \mu)$. Find a formula for M_c .

- i) Using the information obtained above, relate the orbital angular velocity ω to the gravitational-wave frequency f_{GW} . Given that, for any smooth function $F(t)$ and $a \neq 1$:

$$\frac{dF(t)}{dt} = \chi F(t)^a \implies F(t)^{1-a} = \chi(1-a)(t_0 - t),$$

where χ is a constant and t_0 is an integration constant, show that the equation of the previous item implies that the gravitational-wave frequency is:

$$f_{\text{GW}}^{-8/3} = 8\pi^{8/3} \xi \left(\frac{GM_c}{c^3}\right)^{(2/3)+p} (t_0 - t)^{2-p},$$

and determine the constant p .

On September 14, 2015, the event GW150914 was recorded by the LIGO detectors, consisting of two L-shaped arms, each 4 km long. These arms changed their relative length according to the figure below. The detector arms respond linearly to a passing gravitational wave, and the response pattern mimics the wave. This wave was created by two black holes on nearly circular orbits; the loss of energy by gravitational radiation caused the orbit to shrink and, eventually, the black holes to collide. The collision point corresponds, approximately, to the peak of the signal after point D in the figure below.

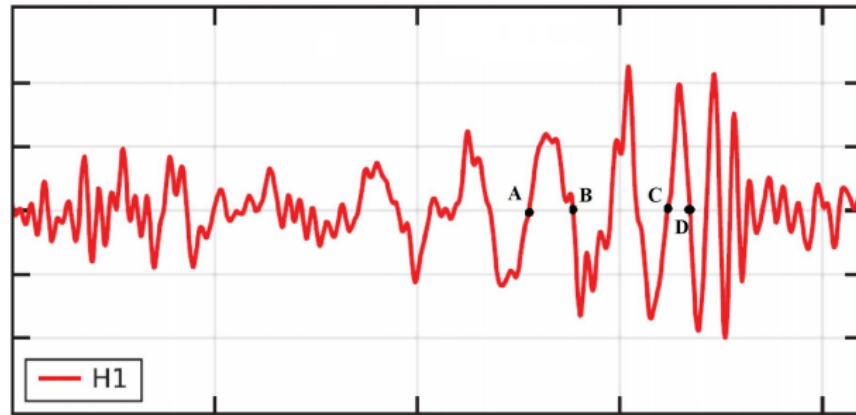


Figure 13: Strain, that is, the relative change in the length of each arm, in the LIGO H1 detector. The horizontal axis represents time, and the points A, B, C, and D correspond to $t = 0.000$, 0.009 , 0.034 , and 0.040 seconds, respectively.

- j) From the figure, estimate $f_{\text{GW}}(t)$ at:

$$t_{AB} = \frac{t_B + t_A}{2} \quad \text{and} \quad t_{CD} = \frac{t_D + t_C}{2}.$$

Assuming that the equation for f_{GW} is valid up to the collision (which, strictly speaking, is not true) and that the two objects have equal masses, estimate the chirp mass, M_c , and the total mass of the system, in terms of solar masses $M_\odot \simeq 2 \times 10^{30}$ kg.

- k) Estimate the minimum orbital separation between the two objects at t_{CD} . Then estimate a maximum size for each object, $R_{\max}$. Obtain $\frac{R_\odot}{R_{\max}}$ in order to compare this size with the radius of our Sun, $R_\odot \simeq 7 \times 10^5$ km. Also estimate their linear orbital speed at the same instant, v_{col} , comparing it with the speed of light, $\frac{v_{\text{col}}}{c}$.

Solution

- a) From Newton's second law,

$$-\frac{GM_1M_2}{(r_1 + r_2)^3} \mathbf{d} = M_1 \mathbf{a}_1$$

where $\mathbf{d}$ is a vector that leaves star two and goes to star one. Therefore, $|\mathbf{d}| = r_1 + r_2$. Using the center-of-mass theorem, we have

$$r_2 = \frac{M_1 r_1}{M_2}$$

Substituting and simplifying,

$$\mathbf{a}_1 = -\frac{GM_2}{r_1^3 \left(1 + \frac{M_1}{M_2}\right)^3} \mathbf{d}$$

Now working with the vector $\mathbf{d}$, note that its direction is the same as that of $\mathbf{r}_1$, since the line joining 1 and 2 necessarily passes through the CM of the system. Thus,

$$\mathbf{d} = |\mathbf{d}| \hat{\mathbf{r}}_1 = (r_1 + r_2) \hat{\mathbf{r}}_1 = (1 + M_1/M_2) \mathbf{r}_1$$

Substituting,

$$\mathbf{a}_1 = -\frac{GM_2}{(1 + M_1/M_2)^2} \frac{\mathbf{r}_1}{r_1^3}$$

With this, we can conclude that

$$\alpha = \frac{GM_2}{(1 + M_1/M_2)^2}, \quad n = 3$$

- b) The energy of the system is given by

$$E = \frac{M_1 v_1^2 + M_2 v_2^2}{2} - \frac{GM_1 M_2}{L}$$

Focusing on the first term and using $v_A = r_A \omega$, we have

$$M_1 v_1^2 + M_2 v_2^2 = (M_1 r_1^2 + M_2 r_2^2) \omega^2$$

By the center-of-mass theorem

$$(M_1 r_1 - M_2 r_2)^2 = 0 \rightarrow M_1 r_1^2 + M_2 r_2^2 = \frac{M_1 M_2}{M_1 + M_2} L^2 \equiv \mu L^2$$

For the second term, it suffices to multiply the numerator and the denominator by $(M_1 + M_2)$. Thus,

$$E = \frac{\mu L^2}{2} - \frac{G}{L} \frac{M_1 M_2}{M_1 + M_2} (M_1 + M_2) \equiv \frac{\mu L^2 \omega^2}{2} + \frac{GM\mu}{L}$$

Thus, $\boxed{A = \frac{\mu L^2 \omega^2}{2}}$.

c) By Kepler's third law,

$$\omega^2 = \frac{GM}{L^3}$$

Substituting,

$$E = \frac{GM\mu}{2L} - \frac{GM\mu}{L} = -\frac{GM\mu}{L}$$

thus concluding that $\boxed{\beta = -1/2}$.

d) Let θ be the angle that the vector r_1 makes with the x -axis. Thus,

$$(x_1, y_1) = r_1(\cos \theta, \sin \theta)$$

The vector r_2 must make an angle of $\pi + \theta$. Using $\sin(\pi + \theta) = -\sin(\theta)$ and $\cos(\pi + \theta) = -\cos \theta$,

$$(x_2, y_2) = -r_2(\cos \theta, \sin \theta)$$

Defining t as the time elapsed since the stars crossed the x -axis, one has $\theta = \omega t$. Therefore,

$$\boxed{(x_1, y_1) = r_1(\cos(\omega t), \sin(\omega t)) \quad (x_2, y_2) = -r_2(\cos(\omega t), \sin(\omega t))}$$

e) The matrix will have the following form,

$$Q = \begin{pmatrix} Q_{11} & Q_{12} & Q_{13} \\ Q_{21} & Q_{22} & Q_{23} \\ Q_{31} & Q_{32} & Q_{33} \end{pmatrix}$$

Invoking the formulas from the statement,

$$Q_{11} = \sum_{A=1}^2 \frac{M_A}{3} (2x_A^2 - y_A^2),$$

$$Q_{22} = \sum_{A=1}^2 \frac{M_A}{3} (2y_A^2 - x_A^2),$$

$$Q_{33} = - \sum_{A=1}^2 \frac{M_A}{3} (x_A^2 + y_A^2),$$

$$Q_{12} = Q_{21} = \sum_{A=1}^2 M_A x_A y_A.$$

For Q_{11} , we have

$$Q_{11} = \frac{M_1}{3} (2x_1^2 - y_1^2) + \frac{M_2}{3} (2x_2^2 - y_2^2)$$

Substituting x and y ,

$$Q_{11} = \frac{M_1}{3} r_1^2 (2 \cos^2(\omega t) - \sin^2(\omega t)) + \frac{M_2 r_2^2}{3} (2 \cos^2(\omega t) - \sin^2(\omega t))$$

$$Q_{11} = \frac{1}{3} (2 \cos^2(\omega t) - \sin^2(\omega t)) (M_1 r_1^2 + M_2 r_2^2)$$

$$Q_{11} = \frac{\mu L^2}{3} (2 \cos^2(\omega t) - \sin^2(\omega t))$$

For Q_{22} ,

$$Q_{22} = \frac{M_1 r_1^2}{3} (2 \sin^2(\omega t) - \cos^2(\omega t)) + \frac{M_2 r_2^2}{3} (2 \sin^2(\omega t) - \cos^2(\omega t))$$

$$Q_{22} = \frac{\mu L^2}{3} (2 \sin^2(\omega t) - \cos^2(\omega t))$$

For Q_{33}

$$Q_{33} = - \frac{M_1 r_1^2}{3} (\cos^2(\omega t) + \sin^2(\omega t)) - \frac{M_2 r_2^2}{3} (\cos^2(\omega t) + \sin^2(\omega t))$$

$$Q_{33} = - \frac{\mu L^2}{3}$$

Finally, for $Q_{12} = Q_{21}$,

$$Q_{12} = Q_{21} = M_1 r_1^2 \cos(\omega t) \sin(\omega t) + M_2 r_2^2 \cos(\omega t) \sin(\omega t)$$

$$Q_{12} = Q_{21} = \mu L^2 \sin(\omega t) \cos(\omega t) = \frac{\mu L^2}{2} \sin(2\omega t)$$

Since the remaining terms are 0, we can choose as A the value $\mu L^2/2$, so our matrix Q takes the form

$$Q = \frac{\mu L^2}{2} \begin{pmatrix} \frac{4}{3} \cos^2(\omega t) - \frac{2}{3} \sin^2(\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & \frac{4}{3} \sin^2(\omega t) - \frac{2}{3} \cos^2(\omega t) & 0 \\ 0 & 0 & -\frac{2}{3} \end{pmatrix}$$

For the terms a_{11} and a_{22} , we will use trigonometric identities, starting with a_{11} ,

$$\frac{4}{3} \cos^2(\omega t) - \frac{2}{3} \sin^2(\omega t) = \frac{2}{3} (2 \cos^2(\omega t) - \sin^2(\omega t))$$

Using $\cos^2(\omega t) = 1 - \sin^2(\omega t)$, we have

$$= \frac{2}{3} (2 - 3 \sin^2(\omega t))$$

Using the double-angle form, we have

$$\cos(2x) = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x \rightarrow \sin^2 x = \frac{1 - \cos(2x)}{2}$$

Making this substitution,

$$= \frac{2}{3} \left(2 - \frac{3}{2} + \frac{3}{2} \cos(2\omega t) \right) = \frac{1}{3} + \cos(2\omega t)$$

Using the same process for a_{22} , we find

$$a_{22} = \frac{1}{3} - \cos(2\omega t)$$

so that our matrix can be simplified to

$$Q = \begin{pmatrix} \frac{1}{3} + \cos(2\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & \frac{1}{3} - \cos(2\omega t) & 0 \\ 0 & 0 & -\frac{2}{3} \end{pmatrix}$$

f) Analyzing the previous matrix, we have

$$A = \frac{\mu L^2}{2}$$

and

$$b_1 = b_2 = \frac{1}{3}, \quad b_3 = -\frac{2}{3}, \quad j_1 = 1, \quad j_2 = -1, \quad j_3 = 0, \quad p_{12} = p_{21} = 2, \quad k = 2$$

g) The formula provided for the power was

$$\mathcal{P} = \frac{G}{5c^5} \sum_{i=1}^3 \sum_{j=1}^3 \left(\frac{d^3 Q_{ij}}{dt^3} \right) \left(\frac{d^3 Q_{ij}}{dt^3} \right).$$

To begin, let us differentiate the matrix Q with respect to time,

$$\frac{d^3 Q}{dt^3} = \frac{\mu L^2}{2} \frac{d^3}{dt^3} \begin{pmatrix} \frac{1}{3} + \cos(2\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & \frac{1}{3} - \cos(2\omega t) & 0 \\ 0 & 0 & -\frac{2}{3} \end{pmatrix}$$

To differentiate a matrix, it suffices to differentiate all of its elements. Doing this, we arrive at

$$\frac{d^3 Q}{dt^3} = \frac{\mu L^2}{2} \begin{pmatrix} 8\omega^3 \sin(2\omega t) & -8\omega^3 \cos(2\omega t) & 0 \\ -8\omega^3 \cos(\omega t) & -8\omega^3 \sin(\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Taking that factor outside the matrix,

$$\frac{d^3 Q}{dt^3} = 4\mu\omega^3 L^2 \begin{pmatrix} \sin(2\omega t) & -\cos(2\omega t) & 0 \\ -\cos(2\omega t) & -\sin(\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

It was given as a hint that the term

$$\sum_{i=1}^3 \sum_{j=1}^3 \left(\frac{d^3 Q_{ij}}{dt^3} \right) \left(\frac{d^3 Q_{ij}}{dt^3} \right)$$

corresponds to the sum of the squares of the entries of the matrix, so

$$\sum_{i=1}^3 \sum_{j=1}^3 \left(\frac{d^3 Q_{ij}}{dt^3} \right) \left(\frac{d^3 Q_{ij}}{dt^3} \right) = (4\mu\omega^3 L^2)^2 (\sin^2(2\omega t) + 2\cos^2(\omega t))$$

Thus,

$$\mathcal{P} = \frac{G}{5c^5} \sum_{i=1}^3 \sum_{j=1}^3 \left(\frac{d^3 Q_{ij}}{dt^3} \right) \left(\frac{d^3 Q_{ij}}{dt^3} \right) = \frac{G}{5c^5} (32\mu^2\omega^6 L^4)$$

Rearranging,

$$\mathcal{P} = \frac{32}{5} \frac{G\mu^2\omega^6 L^4}{c^5}$$

Comparing this expression with the one given in the statement, we have

$$\xi = \frac{32}{5}$$

h) $\mathcal{P}$ represents the power being dissipated by the GWs, so

$$\mathcal{P} = -\frac{dE}{dt}$$

Equating E to the orbital energy,

$$\mathcal{P} = \frac{GM\mu}{2} \frac{d}{dt} \frac{1}{L}$$

We can write ω as a function of L . To do so, it suffices to use Kepler's third law,

$$\omega^2 = \frac{GM}{L^3} \rightarrow \frac{1}{L} = (GM)^{-1/3} \omega^{2/3}$$

Substituting,

$$\begin{aligned} \mathcal{P} &= \frac{(GM)^{2/3} \mu}{2} \frac{d}{dt} \omega^{2/3} = \frac{(GM)^{2/3} \mu}{2} \frac{d}{d\omega} \omega^{2/3} \frac{d\omega}{dt} \\ &= \frac{(GM)^{2/3} \mu}{3\omega^{1/3}} \frac{d\omega}{dt} \end{aligned}$$

Equating this to the expression obtained in the previous item,

$$\frac{\xi G \mu^2 \omega^6 L^4}{c^5} = \frac{(GM)^{2/3} \mu}{3\omega^{1/3}} \frac{d\omega}{dt}$$

Solving for $d\omega/dt$,

$$\frac{d\omega}{dt} = (3\xi) \frac{G \mu \omega^{19/3} L^4}{(GM)^{2/3} c^5}$$

Substituting $L = \left(\frac{GM}{\omega^2}\right)^{1/3}$, we have

$$\frac{d\omega}{dt} = (3\xi) \frac{G \mu \omega^{19/3}}{(GM)^{2/3} c^5} \left(\frac{GM}{\omega^2}\right)^{4/3}$$

Cubing,

$$\left(\frac{d\omega}{dt}\right)^3 = (3\xi)^3 \frac{G^3 \mu^3 \omega^{19}}{G^2 M^2 c^{15}} \frac{G^4 M^4}{\omega^8}$$

Simplifying,

$$\left(\frac{d\omega}{dt}\right)^3 = (3\xi)^3 \frac{\omega^{11}}{c^{15}} (G^5 \mu^3 M^2)$$

Analyzing the expression in the statement, we have

$$M_c = (\mu^3 M^2)^{1/5}$$

i) Using the fact given in the statement,

$$\frac{dF(t)}{dt} = \chi F(t)^a \rightarrow F(t)^{1-a} = \chi(1-a)(t_0 - t)$$

we can find an expression for $\omega(t)$.

$$\frac{d\omega}{dt} = \frac{3\xi(GM_c)^{3/5}}{c^5} \omega^{11/3} \rightarrow \omega^{1-11/3} = \frac{3\xi(GM_c)^{3/5}}{c^5} (1 - 11/3)(t_0 - t)$$

Simplifying,

$$\omega^{-8/3} = -\frac{3\xi(GM_c)^{3/5}}{c^5} \frac{8}{3} (t_0 - t)$$

$$\omega^{-8/3} = \frac{8\xi(GM_c)^{3/5}(t - t_0)}{c^5}$$

To find an expression for f_{GW} , we must use

$$f_{GW} = 2f_{orb} = \frac{\omega}{\pi} \rightarrow \omega = \pi f_{GW}$$

Making this substitution, we arrive at

$$f_{GW}^{-8/3} = \frac{8\pi^{8/3}(GM_c)^{3/5}(t - t_0)}{c^5}$$

Comparing with the form in the statement, we obtain

$$p = 1$$

j) From the figure, we take the two Δt as half-periods. Thus, the cyclic gravitational-wave frequency is $f_{GW} = 1/(2\Delta t)$. The four given points then allow us to calculate the frequency at the mean time of the two intervals as:

t (s)	f_{GW} (Hz)
$t_{AB} = 0.0045$	$f_{GW}(t_{AB}) = (2 \times 0.009)^{-1}$
$t_{CD} = 0.037$	$f_{GW}(t_{CD}) = (2 \times 0.006)^{-1}$

Now we have two pairs of values (f_{GW}, t) for two unknowns (t_0, M_c) . In terms of t_{AB} and t_{CD} , and dividing the two equations, we obtain:

$$t_0 = \frac{At_{CD} - t_{AB}}{A - 1}, \quad A = \left(\frac{f_{GW}(t_{AB})}{f_{GW}(t_{CD})} \right)^{-8/3}.$$

Substituting the numerical values, $A \simeq 2.95$ and $t_0 \simeq 0.054$ s. Now we can use the equation for f_{GW} at either t_{AB} or t_{CD} and determine M_c . One obtains the chirp mass:

$$M_c \simeq 6 \times 10^{31} \text{ kg} \simeq 30 M_\odot$$

Thus, the total mass M is:

$$M = 4^{3/5} M_c \simeq 69 M_\odot$$

This result is, in fact, remarkably close to the best estimates that use the full theory of general relativity! [Even though the real objects do not have exactly equal masses and the theory we have just used is not valid very close to the collision.]

- k) Kepler's law gives $L = (GM/\Omega^2)^{1/3}$. The second pair of points highlighted on the graph corresponds to the cycle before the merger. Thus, using $f_{GW} = \omega/\pi$,

$$\omega_{t_{CD}} \sim 2.6 \times 10^2 \text{ rad/s.}$$

Next, using the total mass obtained previously:

$$L \sim 5 \times 10^2 \text{ km}$$

Thus, these objects have a maximum radius of $R_{\max} \sim 250$ km. Therefore, they have more than 30 times as much mass, and:

$$\frac{R_\odot}{R_{\max}} \sim 3 \times 10^3,$$

they are 3000 times smaller than the Sun!

The linear speed is:

$$v_{\text{col}} = \frac{L}{2} \omega \simeq 7 \times 10^4 \text{ km/s.}$$

They are moving at more than 20% of the speed of light!

$$\boxed{\begin{aligned} L_{\text{col}} &\sim 5 \times 10^2 \text{ km,} \\ \frac{R_\odot}{R_{\max}} &\sim 3 \times 10^3, \\ \frac{v_{\text{col}}}{c} &\sim 0.2. \end{aligned}}$$

- [3] **Problem 17.** (Barra 2024) During his stay at Hotel Fazenda Ribeirão, Hugo was observing the star Plo I with his telescope. Comparing its spectrum with those of other stars, he discovers that Plo I is a neutron star and estimates its mass to be $M_1 = 2M_\odot$. He also observes that Plo I has sinusoidal variations in its radial velocity over time and concludes that Plo I must be part of a binary system with another celestial body, Plo II. Observing the velocity curve of Plo I, Hugo also determines that Plo I has a circular orbit of period P and maximum radial velocity K . However, he does not know the mass M_2 of Plo II, nor the inclination i of the binary's orbital plane with respect to the plane of the sky, shown in the following figure.

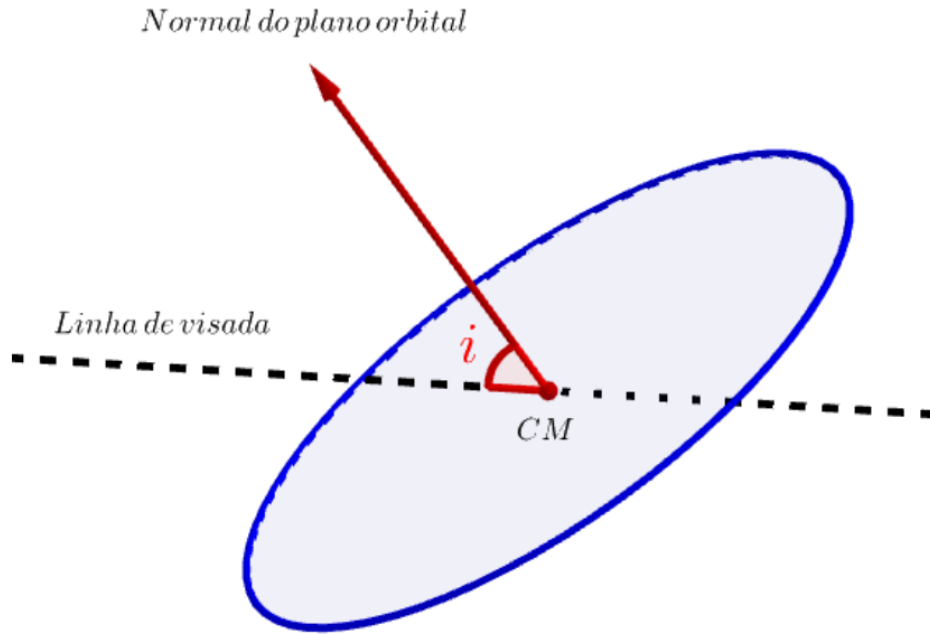


Figure 14: Representation of the orbit

- a) From Kepler's third law and Newton's second law, find an expression for the **mass function** of the binary system formed by Plo I and Plo II in terms of P , K , and universal constants. You may use that the mass function is given by:

$$f = \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2}$$

- b) Using his measurements, Hugo calculates that the mass function of the binary is $f = 46.2M_\odot$. We wish to find the minimum value $M_{2,\min}$ of the mass of Plo II, knowing that it corresponds to the case $i = 90^\circ$. Assume $M_{2,\min} > M_1$. From this, determine $M_{2,\min}$. Is your result consistent with the hypothesis? Answer YES or NO, justifying with calculations.
- c) Based on your answer to the previous item, conclude: is it more likely that Plo II is an exoplanet, a white dwarf, a neutron star, or a black hole?

Solution

- a) First, note that $v_{r, \text{ apparent}} = |\omega \times \mathbf{r}| = \omega r \sin i = v_r \sin i$, so $K = v_r \sin i = \frac{2\pi a_1}{T} \sin i$. Defining $a = a_1 + a_2$ and $M = M_1 + M_2$, where a_i and M_i represent the distance between star i and the CM and the mass of star i , we have Kepler's third law as

$$\frac{a^3}{T^2} = \frac{GM}{4\pi^2}$$

$$a^3 = \frac{GM}{4\pi^2} T^2 \rightarrow T = 2\pi \sqrt{\frac{a^3}{GM}}$$

Substituting $T = \frac{2\pi a_1}{v_1}$

$$(a_1 + a_2)^3 = \frac{G(M_1 + M_2)a_1^2}{v_1^2}$$

$$(1 + a_2/a_1)^2 = \frac{G(M_1 + M_2)}{(a_1 + a_2)v_1^2}$$

Also, using the center-of-mass theorem,

$$\frac{a_1}{a_2} = \frac{M_2}{M_1}$$

Working with the mass function,

$$\frac{1}{f} = \frac{(1 + M_1/M_2)^2}{M_2 \sin^3 i} = \frac{(1 + a_2/a_1)^2}{M_2 \sin^3 i}$$

Substituting $(1 + a_2/a_1)$

$$\frac{1}{f} = \frac{G(M_1 + M_2)}{(a_1 + a_2)M_2 v_1^2 \sin^3 i} = \frac{G(1 + M_1/M_2)}{(a_1 + a_2)v_1^2 \sin^3 i} = \frac{G(1 + a_2/a_1)}{(a_1 + a_2)v_1^2 \sin^3 i}$$

Substituting $(1 + a_2/a_1)$ again,

$$\frac{1}{f} = \frac{G}{v_1^3 \sin^3 i} \sqrt{\frac{G(M_1 + M_2)}{(a_1 + a_2)^3}}$$

But look at the similarity between the term inside the square root and Kepler's third law! Thus,

$$\frac{1}{f} = \frac{2\pi G}{T v_1^3 \sin^2 i}$$

Finally, substituting K and rearranging,

$$f = \frac{TK^3}{2\pi G}$$

b) Assuming $i = 90^\circ$, we have

$$f = \frac{M_2^3}{(M_1 + M_2)^2}$$

Using units of solar masses and $M_1 = 2M_\odot$, we have

$$46.2 = \frac{M_2^3}{(2 + M_2)^2}$$

$$M_2 = (46.2(2 + M_2)^2)^{1/3}$$

Using iteration we can find $M_2 = 49.7M_\odot$

c) Since the mass of M_2 is very large, it is more likely that it is a black hole.

[3] **Problem 18.** Juventino is located at his (not so) secret base at the terrestrial North Pole. After an emergency call from Deduardo Letodo, he needs to send a missile to the equator in order to blow up the base of his archenemy: Raposo.

- a) Juventino is not in his best form, and with all his strength he can only throw the rocket at the first cosmic velocity (the orbital speed for bodies near Earth's surface). Under these conditions, what is the semimajor axis of the orbit of the rocket launched by Juventino?
- b) What is the greatest height the rocket reaches?
- c) What is the time interval between Juventino launching the rocket and Raposo's base being blown up?

Solution

a) Using conservation of energy,

$$-\frac{GM_\oplus m}{R_\oplus} + \frac{mv_0^2}{2} = -\frac{GM_\oplus m}{2a}$$

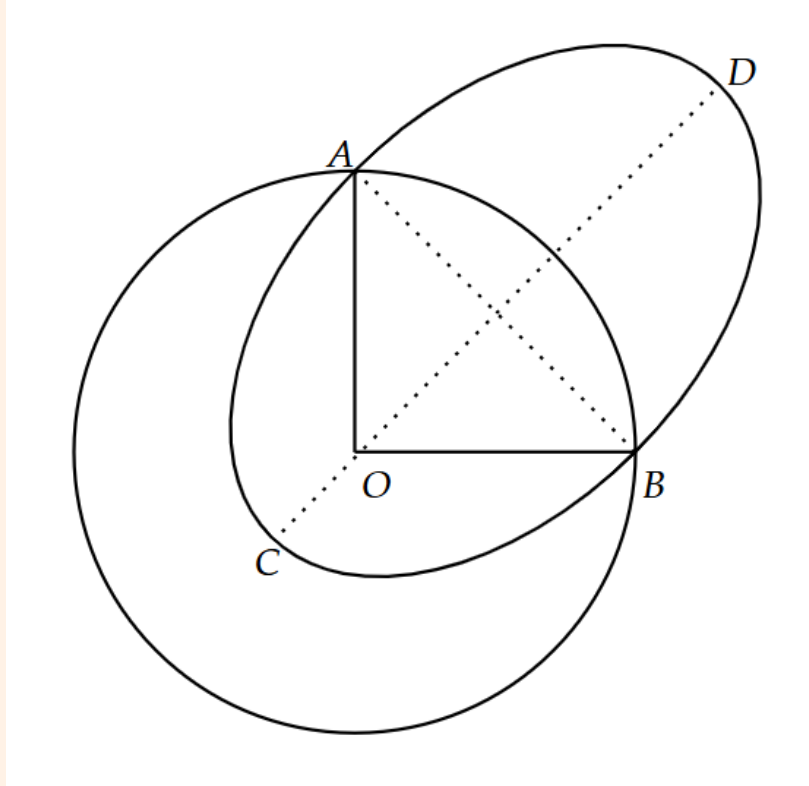
Solving for a ,

$$a = \frac{GM_\oplus}{2GM_\oplus - v_0^2 R_\oplus} R_\oplus$$

Substituting $v_0^2 = \frac{GM_\oplus}{R_\oplus}$, we obtain

$$a = R_\oplus$$

b) Sketching the rocket's orbit,



By geometry, we can say that $\overline{AO} = \overline{BO} = \frac{\overline{CD}}{2} = R_{\oplus}$. Using basic trigonometry, we obtain $\overline{AB} = R_{\oplus}\sqrt{2}$. The greatest height the rocket reaches is given by $\overline{OD} - R_{\oplus}$.

Using the property of the ellipse that the sum of the distances from a point to the two foci is $2a$. Since O is one of the foci, the other focus, O' , is also a distance $R_{\oplus}$ from A and from B . Under these conditions, the point O' must lie a distance $R_{\oplus}\sqrt{2}$ from O and on the line $\overline{CD}$. Thus, $AOBO'$ forms a square of side $R_{\oplus}$.

The midpoint of the segment $\overline{OO'}$ is the center of the ellipse. Therefore, $\overline{AB} = 2b = 2a\sqrt{1 - e^2}$. Substituting the value of $\overline{AB}$, we have

$$R_{\oplus}\sqrt{2} = 2R_{\oplus}\sqrt{1 - e^2}$$

Solving for e ,

$$e = \frac{1}{\sqrt{2}}$$

Therefore, $\overline{OO'} = 2c = 2R_{\oplus}/\sqrt{2}$. Let $\overline{OC} = x$. By symmetry, $\overline{OC} = \overline{O'D}$. Setting up the equation,

$$\overline{CD} = \overline{CO} + \overline{OO'} + \overline{O'D} = 2x + \frac{2}{\sqrt{2}}R_{\oplus} = 2R_{\oplus}$$

$$x = R_{\oplus} \left(1 - \frac{1}{\sqrt{2}} \right)$$

Now we can use $\overline{OD} = \overline{CD} - \overline{OC} = R_{\oplus} (1 + 1/\sqrt{2})$, and therefore the greatest height the rocket reaches is

$$h = \overline{OD} - R_{\oplus} = \frac{R_{\oplus}}{\sqrt{2}}$$

- c) The point A has true anomaly $\theta_A = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$. By symmetry, the time to go from C to A is the same as the time to go from B to C . Therefore, the time to go from A to B is the orbital period minus twice the time from O to A . To calculate this last quantity, we will use Kepler's equation,

$$M = E - e \sin E$$

where E is the eccentric anomaly, defined by

$$r(E) = a(1 - e \cos E)$$

$$E = \cos^{-1} \left(\frac{1 - r/a}{e} \right)$$

Substituting $a = R_{\oplus}$, we can find E_A . When the rocket is at the point A , $r(A) = R_{\oplus}$, so

$$E_A = \frac{\pi}{2}$$

Thus,

$$M_A = \frac{\pi}{2} - \frac{1}{\sqrt{2}}$$

Finally, the time to go from A to B is

$$t_{AB} = T - 2t_{CA} = T \left(1 - \frac{M_A}{\pi} \right)$$

$$t_{AB} = T \left(1 - \frac{1}{2} + \frac{1}{\pi\sqrt{2}} \right)$$

Using Kepler's third law,

$$t_{AB} = \left(\frac{1}{2} + \frac{1}{\pi\sqrt{2}} \right) \sqrt{\frac{4\pi^2 R_{\oplus}^3}{GM_{\oplus}}}$$

$$t_{AB} \approx 1^h 01^m$$

- [5] **Problem 19.** In this problem, our goal is to calculate the precession period of the Earth. To do so, we will consider the Earth as an ellipsoid, with polar radius R_p , equatorial radius R_e , rotation period about the polar axis $T = 23^h 56^m 4^s$, and uniformly distributed mass $M_\oplus$. For our model, consider that only the Sun and the Moon exert gravitational forces relevant to the problem.

- a) Assuming that $R_p = R_\oplus$, what is the numerical value of R_e ?
- b) The moment of inertia of a body can be calculated by $I = \int r^2 dm$. Given this, calculate the moment of inertia about the x -axis and about the y -axis of an ellipsoid given by the equation

$$\frac{x^2}{R_e^2} + \frac{y^2}{R_p^2} + \frac{z^2}{R_p^2} = 1$$

With the moments of inertia calculated, for the Earth let us define the x -axis as the axis that contains the equator and the y -axis as the one that contains the poles. Calculate the quantity

$$\frac{I_p}{(I_p - I_e)}$$

where I_p is the polar moment of inertia (about the y -axis) and I_e the equatorial moment of inertia (about the x -axis)

- c) Find an expression for the torque exerted by the Sun-Moon system on the Earth. (Hint: because of symmetry, many terms inside the integrals will vanish, and the final torque will have a component in only one direction in the coordinate system defined previously.)
- d) Finally, using the equation

$$\tau = \Omega \times \mathbf{L}$$

where $\mathbf{L}$ is the angular momentum and Ω is the precession frequency, find the precession period of the Earth.

Solution

- a) Since the Earth is in hydrostatic equilibrium,

$$\frac{GM_\oplus}{r_p} = \frac{GM_\oplus}{r_e} + \frac{1}{2}\omega^2 r_e^2$$

Solving for r_e ,

$$r_e = \frac{T}{2\pi} \sqrt{2GM_\oplus \left(\frac{1}{r_p} - \frac{1}{r_e} \right)}$$

Iterating, we can obtain

$$r_e \approx 6382 \text{ km}$$

- b) Given the definition of the ellipsoid, x and y lie in the equatorial plane and z passes through the center of the Earth and the north celestial pole. Calculating the moments

of inertia,

$$I_{zz} = \int (y^2 + z^2) dm = \rho \iiint (x^2 + y^2) dx dy dz$$

Using the transformation,

$$x = R_e u \quad y = R_e v \quad z = R_p w$$

so that $u^2 + v^2 + w^2 = 1$, we have

$$I_{zz} = \rho \iiint (R_e^2 u^2 + R_e^2 v^2) R_e^2 R_p du dv dw = \rho R_e^4 R_p \iiint (u^2 + v^2) du dv dw$$

Since $u^2 + v^2 + w^2 = 1$ describes a unit sphere, we have symmetry, which guarantees

$$\iiint u^2 d\Omega = \iiint v^2 d\Omega = \iiint w^2 d\Omega = \frac{1}{3} \iiint (u^2 + v^2 + w^2) d\Omega$$

We can simplify

$$\frac{1}{3} \iiint (u^2 + v^2 + w^2) d\Omega = \frac{1}{3} \int_0^{2\pi} \int_0^\pi \int_0^1 r^4 \sin \theta dr d\theta d\phi = \frac{4\pi}{15}$$

Solving for I_{zz} ,

$$I_{zz} = \rho R_e^4 R_p \left(\frac{4\pi}{15} + \frac{4\pi}{15} \right)$$

Substituting $\rho = 3M_\oplus / 4\pi R_e^2 R_p$, we obtain

$$I_{zz} = \frac{2}{5} M_\oplus R_e^2$$

Doing the same for

$$I_{xx} = \rho R_e^2 R_p \iiint ((R_e v)^2 + (R_p w)^2) du dv dw$$

we obtain

$$I_{xx} = \frac{1}{5} M_\oplus (R_e^2 + R_p^2)$$

Also note that $I_{zz} \equiv I_p$ and $I_{xx} \equiv I_e$. Thus,

$$\boxed{\frac{I_e}{I_p - I_e} \approx 580}$$

Since our value is 93% larger than expected, in reality this quantity must correspond to ≈ 300 . Let us call this quantity f and proceed with it from here on.

c) Since the torque is such that

$$\tau \propto \frac{M}{r^3}$$

we have

$$\frac{\tau_{\text{Moon}}}{\tau_{\odot}} = \frac{M_{\text{Moon}}}{M_{\odot}} \frac{d_{\odot}}{d_{\text{Moon}}} \approx 2.17$$

Now, to find the torque due to the Sun, we first calculate the force it exerts on a mass element dm in the Earth. Let us define $\mathbf{R}$ as the position vector of the Sun and $\mathbf{r}$ as the position vector of the mass element.

$$d\mathbf{F} = -\frac{GM_{\odot}}{|\mathbf{R} - \mathbf{r}|^3}(\mathbf{R} - \mathbf{r})dm$$

With respect to the center of the Earth, the torque is

$$d\vec{\tau} = \mathbf{r} \times d\mathbf{F} = \frac{GM_{\odot}}{|\mathbf{R} - \mathbf{r}|^3}(\mathbf{R} \times \mathbf{r})dm$$

Using $R \gg r$ and the approximation $(1 + x)^n \approx 1 + nx$, we obtain

$$d\vec{\tau} \approx \frac{GM_{\oplus}}{R^3} \left(1 + 3\frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) (\mathbf{R} \times \mathbf{r})dm$$

Expanding the cross product,

$$\mathbf{R} \times \mathbf{r} = (R_y r_z - R_z r_y)\hat{\mathbf{x}} + (R_z r_x - R_x r_z)\hat{\mathbf{y}} + (R_x r_y - R_y r_x)\hat{\mathbf{z}}$$

In this way,

$$\tau_x = \frac{GM_{\oplus}}{R^3} \int \left(1 + 3\frac{R_x r_x + R_y r_y + R_z r_z}{R^2} \right) (R_y r_z - R_z r_y)dm$$

However, when we evaluate the integral, we can see that all terms of the form $\int r_i dm$ and $\int r_i r_j dm$ vanish, due to symmetry. Therefore our integral reduces to

$$\tau_x = \frac{3GM_{\oplus}}{R^5} R_y R_z \left(\int r_y^2 dm + \int r_z^2 dm \right)$$

The remaining integrals are the moments of inertia about y and z . Similarly, for τ_y and τ_z we obtain

$$\tau_x = \frac{3GM_{\oplus}}{R^5} R_y R_z (I_{yy} - I_{zz})$$

$$\tau_y = \frac{3GM_\oplus}{R^5} R_z R_x (I_{zz} - I_{xx})$$

$$\tau_z = \frac{3GM_\oplus}{R^5} R_x R_y (I_{xx} - I_{yy})$$

We can describe R_i using elliptical coordinates.

$$R_x = R \sin \lambda \cos \epsilon$$

$$R_y = R \cos \lambda$$

$$R_z = R \sin \lambda \sin \epsilon$$

where λ is the ecliptic longitude of the Sun. Thus,

$$\tau_x = \frac{3GM_\oplus}{R^3} \sin \lambda \cos \lambda \sin \epsilon (I_{yy} - I_{zz})$$

$$\tau_y = \frac{3GM_\oplus}{R^3} \sin^2 \lambda \sin \epsilon \cos \epsilon (I_{zz} - I_{xx})$$

$$\tau_z = \frac{3GM_\oplus}{R^3} \sin \lambda \cos \lambda \cos \epsilon (I_{xx} - I_{yy})$$

For the precession effect, what matters is the average value of the torque, but note that

$$\langle \sin \lambda \cos \lambda \rangle = 0$$

. Thus, the only torque that remains is that about the y -axis.

$$\vec{\tau} = \langle \tau_y \rangle \hat{\mathbf{y}} = \frac{3GM_\oplus}{2R^3} \sin \epsilon \cos \epsilon (I_{zz} - I_{xx}) \hat{\mathbf{y}}$$

This is only the contribution of the Sun. Adding that of the Moon, we have

$$\boxed{\vec{\tau}_{total} = \frac{4.75GM_\oplus}{R^3} \sin \epsilon \cos \epsilon (I_p - I_e) \hat{\mathbf{y}}}$$

d) Using the given relation,

$$\vec{\tau}_{total} = \vec{\Omega} \times \vec{L} = \Omega L_\perp$$

The value of $L_\perp$ is $I_p \omega \sin \epsilon$. Thus,

$$\frac{4.75GM_\oplus}{R^3} \sin \epsilon \cos \epsilon (I_p - I_e) = \frac{2\pi}{T} I_p \omega \sin \epsilon$$

Solving for T ,

$$\boxed{T = \frac{2\pi}{4.75} \frac{R^3 \omega}{GM_\oplus \cos \epsilon} \frac{I_p}{I_p - I_e} \approx 25300 \text{ years}}$$

which is an estimate with an error of only 2% :)

[5] **Problem 20.** (Magna notes)